

A zero-free half-plane beyond seven eighths for Dirichlet L -functions and finite-order Hecke L -functions over $\mathbb{Q}(\sqrt{-3})$

Jizhou Guo
mitsuha2021b@gmail.com

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Abstract

Using the written statements of OpenAI's *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane* $\text{Re } s > 7/8$ (30 September 2026) as input, we refine its compensated-probe argument. Every Dirichlet L -function, every finite-order Hecke L -function over $\mathbb{Q}(\sqrt{-3})$, and the Riemann zeta function have no zeros in $\text{Re } s > \sigma_{\dagger}$, where $\sigma_{\dagger} = 0.8749570194\dots$ is the root in $(0.8749570194, 0.8749570195)$ of $7884s^3 - 18819s^2 + 14643s - 3686$. In particular, $\text{Re } s > 43747851/50000000$ is zero-free; principal poles at $s = 1$ are allowed. Optimising the skew parameter gives the quadratic endpoint $(1507 - 2\sqrt{921})/1653$. Extending the plain fourth-moment range from $[3/4, 1]$ to $[37/50, 1]$ gives the cubic endpoint $\sigma_{\dagger}$. Exact algebraic certificates are included as ancillary files. One mean-value hypothesis, stated in Section 11 and not proved here, would give the boundary $328/375$.

1 Introduction

OpenAI's paper [1] proves a common zero-free half-plane for the finite-order Hecke family over the Eisenstein field and transfers it to all Dirichlet L -functions. Its completed cubic reflection, compensated probe, marked inverse moment, and plain fourth moment provide the analytic structure used here. We retain that structure and enlarge its geometric and moment parameters.

Taking the written statements of [1] as input, with the hypotheses specified below, we prove:

Theorem 1.1. *Let $\sigma_{\dagger}$ be the unique root in*

$$\frac{8749570194}{10^{10}} < \sigma_{\dagger} < \frac{8749570195}{10^{10}}$$

of

$$G(s) = 7884s^3 - 18819s^2 + 14643s - 3686. \tag{1.1}$$

Every finite-order Hecke L -function over $F_0 = \mathbb{Q}(\sqrt{-3})$ and every Dirichlet L -function, of every modulus, have no zeros in $\text{Re } s > \sigma_{\dagger}$. This includes $\zeta(s)$. The pole of a principal character at $s = 1$ is allowed.

Corollary 1.2. *The same functions have no zeros in*

$$\text{Re } s > \frac{43747851}{50000000} = \frac{7}{8} - \frac{2149}{50000000}.$$

The boundary $20999/24000 = 7/8 - 1/24000$ of Theorem 7.2 was obtained independently by Z. Gao in a draft dated 7 October 2026 [2], with the same parameters $\ell = 1001/6000$ and $b = 1/8$. Gao also identifies the fixed-skew envelope’s limiting boundary as approximately 0.8749572006 . Here we optimise the skew parameter b , obtaining the endpoint $\sigma_* = (1507 - 2\sqrt{921})/1653$ in Theorem 8.1, and extend the plain fourth-moment range to $37/50 \leq \kappa \leq 1$, obtaining σ_+ in Theorem 10.2. The endpoint σ_* , with the same parameters as in Theorem 8.1, was obtained independently by B. Liu in a note dated 8 October 2026, accompanied by Lean files [3]; that note remarks that enlarging the range $3/4 \leq \kappa \leq 1$ of the moment parameter may give a further improvement of about 4.48×10^{-8} and does not pursue it. Argonaut Math [4] states the boundary $7/8 - 1/4000000$, with $\ell = 1/6 + 10^{-6}$ and $b = 1/8$, also with Lean files, and a Lean formalization by the GitHub user LW56e2 [5] states the boundary $4374785349/5000000000$, a rational number exceeding σ_* by less than 10^{-12} . A model computation in [6], dated 7 October 2026, applies the exponent formulas of [1] outside their stated ranges and indicates a boundary near 0.87496 from a change of geometry alone.

The input is OpenAI’s paper [1], dated 30 September 2026; references to that paper use repository `openai/math` at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Its final theorems—the $7/8$ half-plane for ζ , all Dirichlet L -functions and all finite-order Hecke L -functions over $\mathbb{Q}(\sqrt{-3})$ —are covered by the Lean formalization described in [7], with principal poles excluded. The intermediate statements of [1] used here are taken as written, with their hypotheses listed in Appendix B. In particular, its Theorem 1.1 supplies $\beta^* \leq 7/8$. Part II of [1] is written under the standing supposition $\beta^* > 7/8$; Appendix A proves that the auxiliary results used here do not depend on that supposition, retaining their local hypotheses. We change the geometry of the compensated probe and re-derive the completed norm and low estimate, Euler-factor domain, row count, contours and heights that depend on it. Theorem 9.1 extends the plain fourth-moment estimate of [1, Lemma 18.1], and the subsequent parameter optimisation gives an improvement of $4.298 \dots \times 10^{-5}$ over $7/8$ within OpenAI’s method.

Each stage is a separate theorem: the geometric criterion (Theorem 6.1), the general row count (Theorem 5.1), the rational stages (Theorems 7.1 and 7.2), the quadratic boundary (Theorem 8.1), the extended fourth moment (Theorem 9.1), and the cubic boundary (Theorem 10.2). Section 11 states the limit of the specified envelope model and states one unproved mean-value hypothesis under which the method would give $328/375$.

All half-planes in this paper are strict. Their boundaries are common to the entire character family. Constants, smooth orders, excluded prime sets, and lower thresholds for a fixed target may depend on that target.

2 Conventions and analytic inputs

2.1 Characters, coefficients, and uniformity

Put $\omega = e^{2\pi i/3}$, $\mathcal{O} = \mathbb{Z}[\omega]$, and $\lambda = \sqrt{-3}$. For an element or ideal a , write $q_a = N a$; for a nonzero element put $\alpha(a) = a/|a|$. Ideal indices outside the excluded set use their primary generators, congruent to one modulo 3. Element rows retain all their units and prime powers. These are the conventions of [1, §4.1]. The additive kernel is $e(z) = \exp(2\pi i \operatorname{Tr}_{F_0/\mathbb{Q}}(z/\lambda))$, extended to $\mathbb{C}$ as $\exp(4\pi i \operatorname{Im} z/\sqrt{3})$; with $d\mu(z) = (2/\sqrt{3}) dx dy$ the Eisenstein lattice is self-dual.

For a good prime $\mathfrak{p}$, $\chi_{\mathfrak{p}}(u)$ is the sextic residue symbol, with value zero if $\mathfrak{p} \mid u$. Every integer power, including a zeroth or negative power, retains that zero. Thus $\chi_{\mathfrak{p}}(u)^0 = \mathbf{1}_{\mathfrak{p} \nmid u}$. A character presentation includes its defining modulus and every prescribed zero, even at a redundant prime.

A fixed character is fixed as a presentation, not merely as a primitive inducing character. When a displayed product cancels its unit phases, its natural zeros are still retained.

At a base $Z > 1$, define

$$S_\psi(n; W) = Z^{-n/2} \sum_l \psi(l) W(q_l/Z^n), \quad (2.1)$$

$$M_\psi(r; W) = Z^{-r/2} \sum_l \mu(l) \psi(l) W(q_l/Z^r). \quad (2.2)$$

The inverse sum has squarefree ideal indices through μ . The plain sum has all ideal indices. Both retain the same presentation and any specified masks. The functions W are smooth, supported in fixed compact subintervals of $(0, \infty)$. A pure norm twist and a bounded real norm power are part of the smooth profile. All arithmetic coefficients stay independent of the row unless an input explicitly permits otherwise.

The finite group T presenting the fixed supplementary phases is chosen once, before the target. For a fixed primitive target η , the excluded set S contains the primes over 6, the complete defining-modulus supports of the target and the fixed ray characters, and a specified finite initial segment of primes. Enlarging S does not change T . Write $\Theta = \langle \eta, \hat{T} \rangle$, enlarged to include the supplementary characters appearing in the finite transformations; these already factor through the fixed ray convention. On a physical sixth-power-free row,

$$\mathcal{X}_u = \{\nu(n) \chi_n(u)^\varsigma : \nu \in \Theta, \varsigma \in \{-1, 1\}\}. \quad (2.3)$$

This family is closed under conjugation. Its primitive conductors and deletion radicals have norms $\ll_{\mathcal{A}} q_u$ by [1, §8.1]. If u has a prime outside S with valuation $1, \dots, 5$, each presentation is ramified at that prime and cannot induce a character in Θ . The finitely many rows supported on S are treated by the small-row estimate.

Here and below, $\mathcal{A}$ denotes fixed arithmetic data. All logarithmic lengths lie in prescribed bounded sets. An estimate with loss ε allows every specified $\varepsilon > 0$. Real exponent losses and, when stated, the slot mesh are chosen before the slot count and the arithmetic datum. After those are fixed, finite seminorm orders, height orders, constants, and thresholds may depend on them, but remain uniform over the declared moving labels. No moving puncture is absorbed into $\mathcal{A}$.

2.2 The global boundary and buffered witnesses

We use exactly the family supremum in [1, (2.1)]:

$$\beta^* = \sup(\{1/2\} \cup \{\operatorname{Re} \rho : 1/2 \leq \operatorname{Re} \rho \leq 1, L_{F_0}(\rho, \eta) = 0 \text{ for some primitive finite-order } \eta\}). \quad (2.4)$$

The input [1, Theorem 1.1] gives $\beta^* \leq 7/8$. A pole is not included in this supremum. Euler convergence excludes zeros to the right of one.

For retained rows with $q_u \asymp U = Z^d$, use [1, Lemma 8.1]. Its hypotheses are a fixed family (2.3), $0 < d_{\min} \leq d \leq d_{\max}$, $T_1 = Z^\tau > 2$, $0 < \tau \leq d_{\min}/100$, and $0 < e < 10^{-3}$. It gives finitely many bins (i, a) , with $a \geq 51/100$. On each nonfloor bin,

$$a \leq \beta^*, \quad \delta = 2a - 1 \in (1/50, 3/4], \quad (2.5)$$

and there is an actual zero of a member of $\mathcal{X}_u$. On the common buffered rectangle,

$$|L_{\text{orig}}(v, \psi)| + |L_{\text{orig}}(v, \psi)^{-1}| \ll U^\varepsilon, \quad \operatorname{Re} v \geq a + 6e, \quad |\operatorname{Im} v| \leq (3i + 2)T_1, \quad (2.6)$$

$$|L_{\text{orig}}(1 - a - 6e + it, \psi)| \ll U^{a-1/2+12e+\varepsilon} (3 + T_1)^C, \quad |t| \leq (3i + 2)T_1. \quad (2.7)$$

Every original presentation here induces a nonprincipal primitive character; its finite deleted factors are retained in L_{orig} . The floor $a = 51/100$ uses the trivial row count.

The pointwise input [1, Lemma 8.2] requires bounded nonnegative length ranges, uniformly smooth untwisted annular profiles, pure twist height at most $(3i+1)T_1$, and a single total additional-frequency allowance $T_1/2$ in each L -argument. Its further condition is $(1 + T_1)^{A_A} \leq U^{\varepsilon/10}$ with $e < e_0(\varepsilon)$. Under these conditions it gives

$$|M_\psi(r; W)|^2 \ll U^{\delta r + \varepsilon}, \quad |S_\psi(m; W)|^2 \ll U^{\delta \min(m, 1-m) + \varepsilon}.$$

We impose its height condition only after its fixed height order is known. The pre-saturation ranges may be taken as $0 \leq r, m \leq 22$, as in [1, §11.1].

Then [1, Proposition 8.3], whose hypotheses are exactly the nonfloor bin and those of Lemma 8.2, gives for every $t \in [1, 3/2]$ a common presentation, common pure twist, and two saturated witnesses with

$$r \leq t + O(\varepsilon), \quad r + m \geq t - O(\varepsilon), \quad m \leq \frac{1}{2} + O(\varepsilon), \quad |M_r|^2 \gg U^{\delta r - \varepsilon}, \quad |S_m|^2 \gg U^{\delta m - \varepsilon}. \quad (2.8)$$

Presentation labels lie in a fixed finite set; dyadic pairs have logarithmic cardinality. Rowwise choices of smooth parameters are controlled by [1, Lemma 4.5], not by changing prime arithmetic coefficients. All its support boxes are fixed logarithmically, and one joint Fourier measure is used for each row Hilbert space. Sections 5 and 6 verify the character, supply, and height conditions for our applications.

2.3 Continuation and transfer

Proposition 2.1 (Common-signal continuation). *Fix $\sigma \in (1/2, 1)$ and suppose $\Delta = \beta^* - \sigma > 0$. Let $C(s) = s + c$ with $c \in \mathbb{R}$. Suppose there are target-independent $0 < \omega_0 < \Delta$ and $m_0 > 0$ such that, for each primitive target η , there exist a finite S , a holomorphic H_η on $\text{Re } s > \sigma$ with $|H_\eta - 1| \leq 1/2$, and one function J_η satisfying*

$$\begin{aligned} |J_\eta(Z)| &\ll_\eta Z^{C(\sigma) + \omega_0}, \\ |J_\eta(Z) - f_\eta(Z)| &\ll_\eta Z^{C(\beta^*) - m_0}, \\ f_\eta(Z) &= \frac{1}{2\pi i} \int_{(2)} Z^{C(s)} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_{F_0}^S(s, \eta)} ds. \end{aligned}$$

Then these assumptions contradict $\beta^ > \sigma$.*

Proof. This is [1, Proposition 2.1]; we include its argument to specify the family quantifier. Put $\varepsilon_* = \min(\Delta - \omega_0, m_0) > 0$. The triangle inequality bounds $f_\eta(Z)$ by $Z^{C(\beta^*) - \varepsilon_*}$ at infinity. At zero, move its contour from 2 to any fixed $B > 2$. The reciprocal Euler product is bounded there, H_η is bounded, and the Gaussian makes horizontal joins tend to zero. Thus $f_\eta(Z) \ll_{\eta, B} Z^{B+c}$ for $0 < Z \leq 1$.

Consequently

$$F_\eta(s) = \int_0^\infty f_\eta(Z) Z^{-C(s)} \frac{dZ}{Z}$$

converges absolutely and locally uniformly on $\text{Re } s > \beta^* - \varepsilon_*$ and is holomorphic there. On the line $2 + it$, substituting $Z = e^u$ gives an ordinary Fourier transform. The two endpoint bounds make its inverse integrable. Fourier inversion and the identity theorem therefore identify $F_\eta(s)$ with $e^{(s-5/6)^2} H_\eta(s) / L_{F_0}^S(s, \eta)$ for $\text{Re } s > 1$. Since $\beta^* - \varepsilon_* > \sigma$ and $|H_\eta| \geq 1/2$, this continues the reciprocal

holomorphically to $\operatorname{Re} s > \beta^* - \varepsilon_*$. The number ε_* is common to all targets. The supremum (2.4) supplies a target with a zero to its right; attainment of the supremum is unnecessary. Its reciprocal has a pole there. Deleted Euler factors are nonzero for positive real part, so they cannot remove that pole. This is a contradiction. $\square$

Once the primitive Hecke family has a strict zero-free half-plane with boundary in $[1/2, 1)$, [1, Proposition 11.3] transfers it to all finite-order Hecke and Dirichlet characters. Its hypotheses are just this family assertion and the indicated range of the boundary. For clarity, the norm lift of a Dirichlet character χ gives, after a common finite deletion,

$$L_{F_0}^S(s, \chi \circ N) = L^S(s, \chi) L^S(s, \chi \chi_{-3}).$$

The two factors have no poles in $0 < \operatorname{Re} s < 1$, so a zero gives a Hecke zero. The argument also applies on the line $\operatorname{Re} s = 1$ away from $s = 1$. At 1, possible principal pole cancellation is excluded by

$$L(1, \chi_{-3}) = \int_0^1 \frac{dx}{1+x+x^2} = \frac{\pi}{3\sqrt{3}} > 0.$$

All imprimitive functions differ by finite factors $1 - a_p q_p^{-s}$, $|a_p| \leq 1$, nonzero on $\operatorname{Re} s > 0$. This covers every modulus and the principal character giving ζ .

3 The compensated probe and its low estimate

3.1 One finite physical expression

Use the datum $T, S, b_{\text{aux}}, \xi, \tau_\xi, \Xi, W_0, W_1$ of [1, §6.1], with the original zero-extended coefficients. Here b_{aux} generates the squarefree product of the primes in S ; it is unrelated to the real geometric parameter b below. The annular W_0, W_1 are nonnegative and nonzero. Let $I_{\eta; D}(X, Y, Z)$ be [1, (6.1)] with $\mathbf{1}_{D|cn^3}$ inserted in the completed index $A = cn^3$. The scales X, Y, Z in that identity are arbitrary positive numbers.

Fix positive slot lengths ℓ_i of sum ℓ , put $P_i = Z^{\ell_i}$, and take nonnegative, nonzero smooth W_i on fixed annuli. Their underlying prime windows are disjoint before any ray restriction or deletion. Put

$$\mathcal{P}_i(Z) = \{\mathfrak{p} \notin S : \mathfrak{p} \in 1_T, q_{\mathfrak{p}}/P_i \in \operatorname{supp} W_i\}.$$

For a tuple write $p_J = \prod_{i \in J} \mathfrak{p}_i$. The physical expression is

$$I_{\eta, \text{mod}}(Z) = \sum_{(\mathfrak{p}_i) \in \prod_i \mathcal{P}_i(Z)} \prod_i W_i(q_{\mathfrak{p}_i}/P_i) \sum_{J \subseteq \{1, \dots, K\}} (-1)^{|J|} q_{p_J}^{-3/2} \overline{\eta(p_{J^c})} I_{\eta; p_{J^c}}(X/q_{p_J}, Y/q_{p_J}, Zq_{p_{J^c}}). \quad (3.1)$$

This is the operation in [1, (12.5)] for the new scales. In particular, the marked coefficient is $\overline{\eta(\mathfrak{p})}$, as printed in [1, (12.5)]. The selected primes are distinct, so each local operation is composed once. Both estimates of this paper concern (3.1).

Lemma 3.1 (Elementary norm count). *For $R > 0$, the number of nonzero element rows with $q_u \leq R$ is at most $16R$, and the number of nonzero integral ideals with norm at most R is at most $3R$.*

Proof. If $u = x + y\omega$, then $q_u = x^2 - xy + y^2 \geq (x^2 + y^2)/2$. For $R \geq 1$ the number of lattice points is at most $(2\sqrt{2R} + 1)^2 < 16R$, since $9 + 4\sqrt{2} < 16$. The Euclidean ring $\mathcal{O}$ has six units, and each nonzero ideal has six generators of its norm. Divide by six. For $R < 1$ both nonzero counts vanish. $\square$

Theorem 3.2 (Completed-row and low estimates). *Suppose*

$$\begin{aligned} 0 \leq \ell \leq 1/5, \quad M = l_x + l_y = 1 - \ell, \quad b = l_y - l_x > 0, \\ l_x - \ell > 0, \quad l_y - \ell - 11b/6 > 0. \end{aligned} \quad (3.2)$$

Fix any finite slot system as above. For a rescaled subset J put

$$d_J = \sum_{i \in J} \ell_i, \quad M' = M - 2d_J, \quad \ell' = \ell - d_J.$$

Let B_{m, c_T}^J be the marked completed-row factor in the exact separation of (3.1), for a fixed ray class $c_T \in T$. For every $\varepsilon > 0$,

$$\sum_{0 < q_m \ll Z^{M'}} |B_{m, c_T}^J(Z)|^2 \ll Z^{M' + (5\ell + d_J - 1)_+ / 4 + \varepsilon}, \quad (3.3)$$

$$|I_{\eta, \text{mod}}(Z)| \ll Z^{l_x/2 + b/12 + \varepsilon}. \quad (3.4)$$

The first bound is uniform in the rescaled tuple and the moving labels. Constants and thresholds may depend on the fixed datum, slot system, support boxes, and finite seminorms.

Proof. We rederive the completed norm rather than use the fixed- ℓ conclusion of [1, Lemma 15.1]. The marked completed series is

$$T_D(s_0, \psi) = \sum_{c \text{ sf}, n} \gamma_2(c) \overline{\alpha(cn^3)} \psi(cn^3) \mathbf{1}_{D|cn^3} q_c^{-s_0} q_n^{-3s_0 + 1/2}.$$

The c, n are the original indices outside S and may share primes. Mellin inversion at $s_0 = t + 1/2$ leaves the central coefficient

$$\frac{\gamma_2(c) \overline{\alpha(cn^3)} \psi(cn^3)}{q_c^{1/2} q_n} \mathbf{1}_{p_{J^c} | cn^3} \prod_{i \notin J} \overline{\eta(\mathfrak{p}_i)} W_i(\varrho_i), \quad \varrho_i = q_{\mathfrak{p}_i} / P_i. \quad (3.5)$$

Its Gaussian profile has argument

$$\frac{q_{cn^3}}{Z q_{p_{J^c}}} = \frac{r}{\prod_{i \notin J} \varrho_i}, \quad r = q_{cn^3} / Z^{1 + \ell'}.$$

Write $W_G(y) = (2\sqrt{\pi})^{-1} \exp(-\log^2 y / 4)$. On a whole annulus $r = e^k x$, take the single joint profile

$$w_{k, J}(x, \varrho) = \chi(\log x) W_G\left(\frac{e^k x}{\prod_{i \notin J} \varrho_i}\right) \prod_{i \notin J} W_i(\varrho_i),$$

where χ is a fixed logarithmic partition function. The central coefficient (3.5) is unchanged by the annular center. There is no additional $e^{k/2}$ or $Z^{\ell'/2}$ factor. A separated annular profile $V_*(y)$ is written $y^{1/2} W(y)$, with $W = y^{-1/2} V_*$, to match the normalization in [1, (14.2)].

The decomposition of [1, Lemma 4.6] applies because all slot supports are fixed and the Gaussian annular seminorms have summable exponential weights. Choose its truncation width as $\varepsilon_0/2$.

Retain whole annuli meeting that strip, and enlarge a fixed support constant $C_{\text{supp}} \geq 3^4$ to contain the partition radius and all support-comparison constants. Set $\varepsilon_Z = \log C_{\text{supp}} / \log Z$. Once $\varepsilon_Z \leq \varepsilon_0/2$, every retained annulus has completed length

$$N_* = 1 + \ell' + \theta_N, \quad |\theta_N| \leq \varepsilon_0.$$

The removed whole annuli have arbitrarily small total norm. Apply [1, Lemma 4.5] to the single joint profile and then Minkowski. The row-sector reduction [1, Lemma 5.2] uses $f = 1$ and puncture one; it retains all nonzero element rows, including rows meeting S . The row ball depends only on the already frozen rescaled tuple, and hence is independent of surviving slots and dual variables.

We may therefore apply the assumption-free reflected-energy statement of [1, Lemma 14.3], justified in Appendix A. Adopt its local logarithmic labels $O, H, A_0, N_0, S_0, B_0, z_a, v, \ell_b, e_\lambda$. Their meaning is, respectively: powerful row part (the supported squarefree part is frozen separately); residual row; active non-slot support; squarefree assignment; small Ramanujan or zero-mask terms; cube-only assignment; active slot length; squarefree dual length; cube dual length; and the supported λ -length. Actual supports give

$$\begin{aligned} H &\leq M' - O + \varepsilon_Z, \quad \Delta_H := M' - O - H \geq -\varepsilon_Z, \quad O \geq -\varepsilon_Z, \\ 2A_0 &\leq O + \varepsilon_Z, \quad N_0 \leq A_0, \quad z_a \leq \ell', \quad A_0, N_0, S_0, B_0, z_a \geq 0. \end{aligned} \quad (3.6)$$

The dual cutoff and retained support are

$$\begin{aligned} T_d &= 2H + 2A_0 + 2z_a - 1 - \ell' - \theta_N - N_0 - 3B_0, \\ y_0 &:= v + 3\ell_b + e_\lambda \leq T_d + \varepsilon_0, \quad v, \ell_b, e_\lambda \geq -\varepsilon_Z. \end{aligned}$$

Since $M' + \ell' - 1 = -3d_J$,

$$T_d - (H - 3d_J) = H - M' + 2A_0 + 2(z_a - \ell') - N_0 - 3B_0 - \theta_N \leq 2\varepsilon_Z + |\theta_N|.$$

It follows that $\max(H, v + \ell_b) \leq H + 2\varepsilon_0 + 5\varepsilon_Z$.

The aggregate exponent of [1, (14.14)], already including the count of the fixed row parts, is

$$\begin{aligned} E_{\text{ref}} &= O/2 + \max(H, v + \ell_b) - S_0 - B_0 + z_a - u_0 - \ell_b - 2e_\lambda/3 \\ &\quad - (T_d - y_0)_+/2, \quad u_0 = \min\{v, z_a, (v + z_a)/3\}. \end{aligned} \quad (3.7)$$

In particular its $O/2$ is counted once. If $u_0 = z_a$, (3.6) gives

$$E_{\text{ref}} \leq O/2 + H + 2\varepsilon_0 + \frac{20}{3}\varepsilon_Z \leq M' - \Delta_H + 2\varepsilon_0 + \frac{43}{6}\varepsilon_Z \leq M' + 2\varepsilon_0 + \frac{49}{6}\varepsilon_Z.$$

If $u_0 \neq z_a$, then $u_0 \geq v/2 - \varepsilon_Z/2$, and

$$\begin{aligned} u_0 + \ell_b + 2e_\lambda/3 + (T_d - y_0)_+/2 &\geq y_0/4 + (T_d - y_0)_+/2 - \frac{17}{12}\varepsilon_Z \\ &\geq T_d/4 - \frac{17}{12}\varepsilon_Z. \end{aligned}$$

The last inequality holds on both sides of $y_0 = T_d$. The exact remaining normalization is

$$O/2 + \Delta_H + S_0 + B_0 + T_d/4 = \frac{2M' + 2z_a - 1 - \ell' + 2\Delta_H - \theta_N}{4} + \frac{2A_0 - N_0 + B_0 + 4S_0}{4}.$$

Its second numerator is nonnegative. Hence

$$\begin{aligned} E_{\text{ref}} &\leq M' + \frac{1 + 3\ell' - 2M' - 2\Delta_H + \theta_N}{4} + 2\varepsilon_0 + \frac{77}{12}\varepsilon_Z \\ &\leq M' + \frac{1 + 3\ell' - 2M'}{4} + \frac{9}{4}\varepsilon_0 + \frac{83}{12}\varepsilon_Z. \end{aligned}$$

Both branches decrease with Δ_H , so its lower bound can be used on every actual dyad. Since

$$1 + 3\ell' - 2M' = 5\ell + d_J - 1,$$

their maximum is at most $M' + (5\ell + d_J - 1)_+ / 4 + 9(\varepsilon_0 + \varepsilon_Z)$. Choose $\varepsilon_0 \leq \min(1/100, \varepsilon/100)$ and then the fixed-data threshold so that $\varepsilon_Z \leq \min(\varepsilon/100, \varepsilon_0/2)$. Allocate the remaining losses to Lemma 14.3, sectors, common Fourier measures, and logarithmic dyad counts. This proves (3.3), with no assumption that a formal endpoint dyad occurs.

For the low bound, the whole-index indicator may be retained term by term in the absolutely convergent completed series. The fixed-ray decomposition and both linear separations in [1, (6.2)] therefore give its marked version, with exactly B^J above. For a fixed rescaled tuple put $r_J = q_{p_J}/Z^{d_J}$; this lies in a fixed positive compact interval. Its actual scales are

$$X' = Z^{l_x - d_J} / r_J, \quad Y' = Z^{l_y - d_J} / r_J, \quad Q = q_{b_{\text{aux}}} Z^{M'} / r_J^2, \quad P_a = Y'^2 / Q = q_{b_{\text{aux}}}^{-1} Z^b.$$

They are at least one after a fixed threshold. Moreover

$$l_y - d_J - 11b/6 \geq l_y - \ell - 11b/6 > 0, \quad P_a^2 / Y' \ll P_a^{1/6}.$$

The additive factor is the original $A_{m, c_T, v}(Y')$ of [1, §6.1]; its fixed ray coefficient has not changed. Thus every hypothesis of the assumption-free [1, Proposition 15.2] holds: polynomial $Q, Y' \geq 1$, $P_a \geq 1$, fixed row ball, and its stated ray-coefficient class. Its norm is

$$\sum_{q_m \ll Q} |A_{m, c_T, v}(Y')|^2 \ll (1 + |v|)^{J_{\text{Gr}}(Q/Y')} P_a^{1/6} Z^\varepsilon.$$

Cauchy–Schwarz in the same marked separation, keeping the outside $Q^{-1/2}$, gives for this tuple

$$Z^{(l_x - d_J)/2 + b/12 + (5\ell + d_J - 1)_+ / 8 + \varepsilon}.$$

The fixed Mellin test absorbs its polynomial height factor. Lemma 3.1 bounds the number of rescaled tuples by $O(Z^{d_J + \varepsilon})$, while the outside coefficient in (3.1) is $O(Z^{-3d_J/2})$. The additional exponent is

$$-d_J + (5\ell + d_J - 1)_+ / 8 \leq -7d_J/8 \leq 0,$$

because $\ell \leq 1/5$. Allocate the finitely many small losses before summing the subsets. This proves (3.4). $\square$

4 Euler factors, compensation, and the enlarged domain

4.1 The six local cases

We use the exact rational identities of [1, Lemma 7.1, (7.6), (7.11)–(7.13), (7.17)]. Their hypotheses are a nonzero sixth-power-free physical row u , $(u, S) = 1$, and a prime $\mathfrak{p} \notin S$. Write $Q = q_{\mathfrak{p}}$, $j = v_{\mathfrak{p}}(u) \in \{0, \dots, 5\}$, $t_{\mathfrak{p}} = \eta(\mathfrak{p})$, and

$$\begin{aligned} \rho &= \chi_{\mathfrak{p}}(u/\mathfrak{p}^j), \quad a_{\mathfrak{p}} = \overline{\alpha(\mathfrak{p})}^3 t_{\mathfrak{p}}^3, \quad b_{\mathfrak{p}} = \overline{\alpha(\mathfrak{p})}^2 t_{\mathfrak{p}}^2 \overline{\chi_{\mathfrak{p}}(4)}, \quad b_{\mathfrak{p}}^3 = a_{\mathfrak{p}}^2, \\ V &= Q^{-6z}, \quad R = a_{\mathfrak{p}}^2 Q^{4-6s-6z}, \quad W_{\text{loc}} = \rho Q^{-w}, \\ W &= \chi_{\mathfrak{p}}(u) Q^{-w}, \quad D = t_{\mathfrak{p}} \overline{\chi_{\mathfrak{p}}(u)} Q^{-s}. \end{aligned}$$

The values $\rho, a_{\mathfrak{p}}, b_{\mathfrak{p}}, t_{\mathfrak{p}}$ have absolute value one; this is explicit in [1, (7.6)]. The original zero extension makes $D = W = 0$ when $\mathfrak{p} \nmid u$, while W_{loc} still has its unit phase. Put

$$P_{\mathfrak{p}}^* = \frac{1}{1-R} \left\{ \frac{R(1-Q^{-1}) - t_{\mathfrak{p}}(Q-1)Q^{-s-w}V^{\mathbf{1}_{j \leq 1}}}{1-V} + J_j \right\}, \quad (4.1)$$

where the six values are

j	J_j
0	$-t_{\mathfrak{p}}\rho^{-1}Q^{-s} + W_{\text{loc}}R$
1	$t_{\mathfrak{p}}Q^{-s-w}$
2	$a_{\mathfrak{p}}\rho^{-3}Q^{3/2-3s}$
3	$-t_{\mathfrak{p}}b_{\mathfrak{p}}\rho^{-2}Q^{2-3s-w} + b_{\mathfrak{p}}^2\rho^{-4}Q^{2-4s}$
4	$-t_{\mathfrak{p}}a_{\mathfrak{p}}\rho^{-3}Q^{5/2-4s-w}$
5	$-a_{\mathfrak{p}}^2Q^{3-6s}$.

The unmarked and scalar-corrected factors are

$$P_{\mathfrak{p}} = \frac{1-VW}{(1-V)(1-W)} + P_{\mathfrak{p}}^*, \quad H_{\mathfrak{p}} = \frac{1-VW + (1-V)(1-W)P_{\mathfrak{p}}^*}{1-D}. \quad (4.2)$$

These identities retain all phases and zeros.

Lemma 4.1 (Enlarged Euler domain). *On*

$$\mathcal{D}'_2 = \{\text{Re } s \geq 87/100, \text{ Re } w \geq 19/20, \text{ Re } z \geq 33/200\}, \quad (4.3)$$

$H_{\eta,u} = \prod_{\mathfrak{p} \notin S} H_{\mathfrak{p}}$ is holomorphic in a neighborhood of every point and $|H_{\eta,u}| \ll_{\varepsilon} q_u^{\varepsilon}$, uniformly in all imaginary parts and unit phases. More explicitly,

$$|H_{\mathfrak{p}} - 1| \leq 210Q^{-181/100} \quad (\mathfrak{p} \nmid u), \quad |H_{\mathfrak{p}} - 1| \leq 32Q^{-82/100} \quad (\mathfrak{p} \mid u).$$

If S contains all primes of norm at most $P_0 = 2^{20}$, then

$$\sup_{\mathcal{D}'_2} |H_{\eta,1} - 1| \leq 1/31. \quad (4.4)$$

Proof. Here $|R| \leq Q^{-221/100}$, $|V| \leq Q^{-99/100}$, $|D| \leq Q^{-87/100}$, and $|W| \leq Q^{-19/20}$. Hence $|1-R|^{-1} \leq 2$ and $|1-V|^{-1}, |1-D|^{-1} \leq 3$. For $j = 0$, put $E_{\mathfrak{p}} = P_{\mathfrak{p}}^* + D$. The exact table gives

$$E_{\mathfrak{p}} = \frac{R\{(1-Q^{-1})/(1-V) + W - D\} - t_{\mathfrak{p}}(Q-1)Q^{-s-w}V/(1-V)}{1-R}. \quad (4.5)$$

Thus $|E_{\mathfrak{p}}| \leq 10Q^{-221/100} + 6Q^{-181/100} \leq 16Q^{-181/100}$. Substituting $P_{\mathfrak{p}}^* = -D + E_{\mathfrak{p}}$ in (4.2) gives

$$H_{\mathfrak{p}} - 1 = \frac{D(V+W-VW) - VW + (1-V)(1-W)E_{\mathfrak{p}}}{1-D}.$$

The three numerator bounds and their combined defect estimate are

$$3Q^{-182/100}, \quad Q^{-194/100}, \quad 64Q^{-181/100}, \quad |H_{\mathfrak{p}} - 1| \leq 204Q^{-181/100} < 210Q^{-181/100}.$$

For $j > 0$, $H_{\mathfrak{p}} - 1 = (1-V)P_{\mathfrak{p}}^*$. The strict second-family term, with $V^{\mathbf{1}_{j \leq 1}} = 1$ for $j \geq 2$, has exponent at most $-82/100$. The boundary terms for $j = 1, 2, 3, 4, 5$ have exponents at most

$$-182/100, \quad -111/100, \quad (-156/100, -148/100), \quad -193/100, \quad -222/100.$$

Their coefficients have modulus one, and the R term is $O(Q^{-221/100})$. The denominator bounds yield

$$|P_{\mathfrak{p}}^*| \leq 2(3 + 3 + 2)Q^{-82/100} = 16Q^{-82/100}, \quad |H_{\mathfrak{p}} - 1| \leq 32Q^{-82/100}.$$

Lemma 3.1 proves normal convergence of the unramified product. The strict exponent margin remains valid in a neighborhood of each boundary point. The finite ramified product satisfies

$$\prod_{\mathfrak{p}|u} (1 + 32Q^{-82/100}) \leq C_{\varepsilon} q_u^{\varepsilon}.$$

For example take $R_{\varepsilon} = \max\{3, (32/\varepsilon)^{50/41}\}$ and $C_{\varepsilon} = 33^{3R_{\varepsilon}}$; above R_{ε} , $32Q^{-82/100} \leq \varepsilon \log Q$.

For $u = 1$, partial summation with the ideal count $A(t) \leq 3t$ gives

$$\sum_{q_{\mathfrak{p}} > P_0} q_{\mathfrak{p}}^{-181/100} \leq \frac{181}{27} P_0^{-81/100}.$$

The defect sum is at most $210(181/27)2^{-81/5} < 1408 \cdot 2^{-16} < 1/32$. Since $e^x \leq (1 - x)^{-1}$ for $0 \leq x < 1$, the product differs from one by at most $1/31$. An enlargement of S decreases this positive majorant. $\square$

4.2 The full correction and principal normalizer

Define the local replacement, with the marked conjugation as printed in [1, (16.2)],

$$G_{\mathfrak{p}} = \frac{(\overline{t_{\mathfrak{p}}}Q^s - Q^{-w})(1 - V)(1 - W)P_{\mathfrak{p}}^* - Q^{-w}(1 - VW)}{1 - D}. \quad (4.6)$$

Direct substitution in (4.2) proves

$$\overline{t_{\mathfrak{p}}}Q^{s+z-1}P_{\mathfrak{p}}^* - Q^{z-w-1}P_{\mathfrak{p}} = \frac{1 - D}{(1 - V)(1 - W)}Q^{z-1}G_{\mathfrak{p}}. \quad (4.7)$$

In particular, $\overline{t_{\mathfrak{p}}}Q^s D = \overline{\chi_{\mathfrak{p}}(u)}$. The full correction is the quotient-free expression

$$\mathcal{H}_{\eta,u,Z}(s, w, z) = \sum_{(\mathfrak{p}_i)} \prod_i [W_i(q_{\mathfrak{p}_i}/P_i)q_{\mathfrak{p}_i}^{z-1}G_{\mathfrak{p}_i}] \prod_{\mathfrak{p} \notin S \cup \{\mathfrak{p}_i\}} H_{\mathfrak{p}}. \quad (4.8)$$

We never divide by a selected $P_{\mathfrak{p}}$ or $H_{\mathfrak{p}}$ in this definition.

Proposition 4.2 (Correction identity and normalization). *For every positive geometry and fixed disjoint slot system used here, the exact high representation of (3.1) is [1, (16.7)] with correction (4.8). This correction is holomorphic in neighborhoods of $\mathcal{D}'_2$ and of the original domains*

$$\mathcal{D}_1(\varepsilon_0) = \{s_r \geq 51/100, z_r \geq 17/50, w_r \geq -1/100, s_r + w_r \geq 1 + \varepsilon_0\}, \quad \varepsilon_0 > 0.$$

It has a fixed scale-power bound on each fixed real box, uniform in imaginary parts, for q_u in a fixed polynomial range. Its scale degree is independent of later external derivative orders.

For $u = 1$ and $Q > P_0$, on $\mathcal{D}'_2$,

$$B_{\mathfrak{p}} := G_{\mathfrak{p}}/H_{\mathfrak{p}} = -1 + O(1400Q^{-87/100}), \quad |G_{\mathfrak{p}}| \leq 3.$$

If K is even and

$$S_i(Z) = \sum_{\mathfrak{p} \in \mathcal{P}_i(Z)} W_i(q_{\mathfrak{p}}/P_i)q_{\mathfrak{p}}^{-5/6}, \quad A_T(Z) = Z^{-\ell/6} \prod_i S_i(Z), \quad (4.9)$$

then $A_T(Z) \asymp (\log Z)^{-K} > 0$ eventually, and for $0 < \mu < (87/100) \min_i \ell_i$,

$$\mathcal{H}_{\eta,1,Z}(s, 1, 1/6) = H_\eta(s) Z^{\ell/6} A_T(Z) (1 + R_{\eta,Z}(s)), \quad |R_{\eta,Z}(s)| \ll Z^{-\mu}, \quad (4.10)$$

uniformly on the principal line $\operatorname{Re} s = \beta^* + e$, provided $\beta^* + e \geq 87/100$, where $H_\eta(s) = H_{\eta,1}(s, 1, 1/6)$. In every application below this condition follows from $\beta^* > \sigma_g \geq 1049/1200 > 87/100$. Also

$$|\mathcal{H}_{\eta,1,Z}(s, w, z)| \ll Z^{\ell \operatorname{Re} z + \varepsilon} \quad ((s, w, z) \in \mathcal{D}'_2 \text{ in a fixed real box}). \quad (4.11)$$

Proof. The denominators of (4.6) are only $1 - R, 1 - V, 1 - D$ and do not vanish in these domains. On a fixed real box bounded by $M_0 \geq 1$, the finite local monomials and geometric denominators give $|G_p| \leq CQ^{20M_0+20}$, independent of imaginary parts. The finite tuple count is a fixed scale power. Unselected products are bounded by Lemma 4.1 and the first-domain bound of [1, Lemma 7.1, (7.14)], whose stated real inequalities are exactly those displayed above. This proves the holomorphy and all-height assertion.

On $s = w = 3, z = 2$, the source identity [1, (7.5)] and all local sums converge absolutely. Apply (4.7) once at each distinct selected prime and sum the finite tuples and subsets. This proves the high representation for the same expression (3.1). No relation among the positive scales enters this finite identity.

For $u = 1$, substituting $P_p^* = -D + E_p$ gives

$$B_p + 1 = \frac{(1-V)(1-W)}{(1-D)H_p} \left\{ (1-W) \left(\frac{V}{1-V} - D \right) + \overline{t}_p Q^s E_p + (1-W) E_p \right\}.$$

The prefactor is at most 24. The first bracketed term is at most $8Q^{-87/100}$. Keep the two terms of (4.5) separate when multiplying by Q^{s_r} :

$$Q^{s_r} |E_p| \leq 10Q^{4-5s_r-6z_r} + 6Q^{1-w_r-6z_r} \leq 16Q^{-87/100}.$$

The last term is at most $32Q^{-87/100}$. Thus the error is at most $1344Q^{-87/100} < 1400Q^{-87/100}$. For $Q > 2^{20}$ this is less than one; indeed $1400^{100} < 2^{1740}$. The unramified factor also has modulus at most $3/2$, giving $|G_p| \leq 3$.

The assumption-free fixed-ray prime normalizer [1, Lemma 13.1] applies: T is fixed, W_i is annular and nonnegative, and only a fixed finite set S is removed. It gives

$$S_i(Z) \sim \frac{P_i^{1/6}}{|T| \log P_i} \int_0^\infty W_i(t) t^{-5/6} dt > 0.$$

Factoring $H_{\eta,1}$ from (4.8) is legitimate here since each principal local factor is nonzero. Summing $B_p = -1 + O(Q^{-87/100})$ in each slot proves (4.10); even K removes the sign. The bound $|G_p| \leq 3$, the positive unselected-product majorant, and the ideal count give (4.11), including the whole residue rectangle. This last absolute bound is used before the residue approximation. $\square$

4.3 Central cutoff and outer contours

Lemma 4.3 (Pretarget cutoff and absolute tuple bounds). *For $51/100 \leq a \leq 7/8$, $0 < e < 10^{-3}$, set*

$$s_r = a + 16e, \quad w_r = 1 - a - 6e, \quad z_r = 17/50.$$

Then

$$|H_p - 1| \leq 256Q^{-1-10e} \quad (\mathfrak{p} \nmid u), \quad |H_p - 1| \leq 256Q^{-10e} \quad (\mathfrak{p} \mid u).$$

Consequently the target-independent cutoff

$$P_{\text{pre}}(e) = \max\{2^{20}, 2^{1/e}\} \quad (4.12)$$

makes every central selected $H_{\mathfrak{p}}$ nonzero, with $|H_{\mathfrak{p}}| > 3/4$.

On $s_r \geq 87/100, w_r = 1/2, z_r = 17/50$, one has $G_{\mathfrak{p}} = O(1)$ for $\mathfrak{p} \nmid u$ and $G_{\mathfrak{p}} = O(Q^{1/2})$ for $\mathfrak{p} \mid u$. On $s_r = w_r = 2, z_r = z_{\infty} > 2$, one has $G_{\mathfrak{p}} = O_{z_{\infty}}(1)$ for every selected prime. On both types of contour,

$$\sum_{(\mathfrak{p}_i)} \prod_i [|W_i(q_{\mathfrak{p}_i}/P_i)| q_{\mathfrak{p}_i}^{z_r-1} |G_{\mathfrak{p}_i}|] \prod_{\mathfrak{p} \notin S \cup \{\mathfrak{p}_i\}} |H_{\mathfrak{p}}| \ll_{\varepsilon} q_u^{\varepsilon} Z^{\ell z_r}. \quad (4.13)$$

Proof. On the central line $w_r \geq 119/1000$. The inverse denominators $1-R, 1-V, 1-D$ are bounded by 2, 2, 4. For $j = 0$, (4.5) gives

$$|E_{\mathfrak{p}}| \leq 8Q^{-11/10-96e} + 4Q^{-51/25-10e} \leq 12Q^{-1-10e}.$$

The DW exponent is exactly $-1-10e$ and the other terms in (4.2) are smaller, giving $208Q^{-1-10e}$. For $j > 0$, the strict term is bounded by Q^{-10e} ; the five boundary cases give respectively

$$-1-10e, \quad -3/100-48e, \quad (-1/50-42e, -1/25-64e), \quad -3/100-58e, \quad -3/50-96e.$$

They are at most $-10e$. Thus $|P_{\mathfrak{p}}^*| \leq 12Q^{-10e}$ and $|H_{\mathfrak{p}} - 1| \leq 24Q^{-10e}$. The common constant 256 suffices. For $Q > 2^{1/e}$, $256Q^{-10e} < 256 \cdot 2^{-10} = 1/4$.

The small contour lies in $\mathcal{D}_1(1/3)$ since $87/100 + 1/2 > 4/3$. In the unramified formula for $G_{\mathfrak{p}}$, its four error exponents are at most

$$-87/100, \quad -51/25, \quad -239/100, \quad -77/50.$$

In the ramified cases, multiplication by Q^{s_r} makes the strict exponent $1/2$, while the boundary exponents are at most

$$-1/2, \quad -24/100, \quad (-24/100, -61/100), \quad -61/100, \quad -135/100.$$

The R exponent is at most $-239/100$ and the rescaled term is $O(Q^{-1/2})$. This proves the stated local bounds. At $s_r = w_r = 2, z_r > 2$ all those exponents are negative; the strict one is at most -1 .

In each slot the unramified prime count contributes $O(P_i^{z_r})$. The ramified primes are divisor-many, contributing $O_{\varepsilon}(q_u^{\varepsilon} P_i^{z_r-1/2})$. Allocate the arbitrary small loss among the fixed slots and multiply the positive unselected-product bound. This proves (4.13) without dividing by possibly zero local factors. $\square$

5 Row counts for the general family supremum

Set $\alpha_0 = 5/6$. A main physical prime factor on the central contour is

$$\mathcal{Q}_i(u; z) = P_i^{-1/2} \sum_{\mathfrak{p} \in \mathcal{P}_i(Z)} \overline{\chi_{\mathfrak{p}}(u)} W_i(q_{\mathfrak{p}}/P_i) (q_{\mathfrak{p}}/P_i)^{z-1}.$$

The assumption-free [1, Lemma 19.1] applies under the buffered hypotheses of Lemma 8.2, with the prime-annulus frequency in the same cumulative allowance. It bounds $|\mathcal{Q}_i|$ by $U^{\varepsilon} P_i^{\alpha-1/2+O(e)}$. After the lengths are fixed, choose small e and a fixed amplitude width ϑ to obtain bins with

$$0 \leq g_i \leq \delta/2, \quad |\mathcal{Q}_i| \leq P_i^{g_i+\vartheta}, \quad g_i > 0 \implies |\mathcal{Q}_i| \geq P_i^{g_i}.$$

Set $g_i = 0$ on error slots, and put $q = \sum_i \ell_i g_i / \ell$, $x = q/\delta \in [0, 1/2]$. The letter q here denotes amplitude, not the moving-support exponent in the fourth moment.

Theorem 5.1 (General row count). *Assume $51/100 < \beta^* \leq 7/8$. Use a fixed nonfloor dynamic and amplitude bin of the retained physical family, with its original masks, presentations, and cumulative height allocation. Suppose*

$$37/50 \leq \kappa \leq 3/4, \quad \beta^* \leq (1 + \kappa)/2.$$

Write

$$c_\kappa = \frac{1}{3\kappa}, \quad A_x = 2 - 2c_\kappa x, \quad B_x = 1 - x, \quad D_x = A_x + B_x, \quad P_x = A_x B_x. \quad (5.1)$$

If the total available prime length at base U exceeds $7/37$ by a fixed positive amount, then for every $t \in [1, 3/2]$ and every $\varepsilon > 0$,

$$\#\mathcal{B} \ll U^{\max\{R_{\text{short},\kappa}(t), L(t)\} + \varepsilon} (1 + T_1)^{A_A}, \quad (5.2)$$

where

$$R_{\text{short},\kappa}(t) = 1 - \delta + \frac{\delta P_x}{D_x} (3/2 - t), \quad (5.3)$$

$$L(t) = 1 - \delta + (5/6 - \delta)(t - 1). \quad (5.4)$$

The output loss, capacity decrement, and mesh may be chosen using only the loss and bounded real ranges, before the target and slot count. After the fixed system and datum are chosen, A_A and all internal seminorm orders are finite and independent of any later external tail order. At $\kappa = 3/4$ the displayed hypothesis follows from the input $\beta^ \leq 7/8$.*

Without selected slots, no supply condition is needed and

$$\#\mathcal{B} \ll U^{1-2\delta/3+\varepsilon} (1 + T_1)^{A_A}. \quad (5.5)$$

Proof. The marked inverse moment used here is the assumption-free [1, Lemma 17.1]. At row width one it requires

$$r + 2z \leq 1 - c_1, \quad 2r + 8z \leq 3 - c_2, \quad c_1, c_2 > 0.$$

Its coefficients are one common fixed finite-ray twist times either common orientation of the sextic symbol, with row-independent bounded prime coefficients and disjoint supports. The fourth moment is the assumption-free [1, Lemma 18.1] when $\kappa = 3/4$ and Theorem 9.1 when $\kappa < 3/4$. Its positive-slot premise is exactly the displayed bound on β^* . Theorem 9.1 is proved without this row-count theorem, so this forward reference introduces no circular dependence.

Subdivide (2.8) by presentation and dyadic pair. If needed, conjugate both witnesses and their profiles to write their common character as $\psi_u^-(n) = \nu(n) \overline{\chi_n(u)}$. Relative to it a physical slot has coefficient

$$\overline{\nu(\mathfrak{p})} \mathbf{1}_{\mathfrak{p} \in 1_T} = |T|^{-1} \sum_{\theta \in \hat{T}} (\overline{\nu} \theta)(\mathfrak{p}),$$

which is fixed across rows and belongs to the required finite character span. The inverse moment permits this negative orientation. For the plain moment conjugate the whole product of two plain copies and all selected slots. Its character becomes $\overline{\nu}(n) \chi_n(u)$ and its slot coefficient becomes $\nu(\mathfrak{p}) \mathbf{1}_{\mathfrak{p} \in 1_T}$. Its absolute square, masks, and disjoint windows are unchanged. Every retained nonunit good row induces outside Θ , by the local ramification discussed in Section 2. The finite-parameter Sobolev assertion of [1, Lemma 4.5] handles the rowwise smooth witnesses with a fixed height cost.

At base U let $w_i = \ell_i/d$. Order positive g_i decreasingly and fill a requested length z fractionally. The average gain is at least qz . If positive slots do not fill z , retaining all of them gives at least

$q\ell/d \geq qz$. Remove the one possible fractional slot, and first reduce the requested length by a fixed $\nu_0 > 0$. The squared gain is at least

$$2qz - 2q\nu_0 - \delta \max_i w_i.$$

These two losses can be made smaller than any prescribed positive power.

For $r < 1, m < 1/2$, the ideal capacities are

$$z_I = (1 - r)/2, \quad z_P = (1 - 2m)/(6\kappa).$$

Dividing the moment bounds by the saturated spikes and the slot gain gives the affine counts

$$\begin{aligned} I(r) &= 1 - \delta\{x + (1 - x)r\}, \\ P_t(r) &= 1 - \delta\{c_\kappa x + (2 - 2c_\kappa x)(t - r)\}. \end{aligned}$$

The second uses two copies of S_m and hence divides by $|S_m|^4$; replacing m by $t - r - O(\varepsilon)$ is legitimate because its exponent decreases with m .

The two lines intersect at

$$r_\kappa(t) = \frac{(2 - 2c_\kappa x)t + (c_\kappa - 1)x}{D_x}. \quad (5.6)$$

For $4/9 \leq c_\kappa \leq 50/111$,

$$23/37 \leq r_\kappa(t) \leq 1, \quad 1/3 \leq t - r_\kappa(t) \leq 1/2. \quad (5.7)$$

To verify these bounds, at $t = 1$,

$$r_\kappa(1) = \frac{2 - (1 + c_\kappa)x}{3 - (1 + 2c_\kappa)x}, \quad \partial_{c_\kappa} r_\kappa(1) = \frac{x(1 - x)}{D_x^2} \geq 0,$$

and $r_{3/4}(1) \geq 23/37$. Also

$$1 - r_\kappa(1) - 1/3 = \frac{x(1 - c_\kappa)}{3D_x} \geq 0.$$

At $t = 3/2$ the two lengths are $1, 1/2$; their affine dependence on t proves the remaining bounds.

For $r \leq r_\kappa(t)$ select the plain count; for $r \geq r_\kappa(t)$ select the inverse count. On the latter side, after the decrement,

$$1 - r - 2z \geq 2\nu_0, \quad 3 - 2r - 8z = 4(1 - r - 2z) + (2r - 1) \geq 8\nu_0 + 9/37.$$

These are fixed positive inverse margins. The maximum inverse capacity is $7/37$. The plain capacity on its selected side is at most

$$1/(18\kappa) \leq 25/333 < 7/37,$$

apart from the arbitrarily small witness loss. Thus the supply and mesh hypotheses hold.

The decreasing I and increasing P_t are bounded by their crossing value. The identity

$$x(A_x + c_\kappa B_x) = D_x - 3P_x/2$$

shows that this value is (5.3). If $r \geq 1$, the assumption-free [1, Lemma 17.6] gives unmarked exponent $1/6 + 5r/6$. Dividing by the inverse spike gives $1/6 + (5/6 - \delta)r \leq L(t) + O(\varepsilon)$, since $\delta \leq 3/4 < 5/6$ and $r \leq t + O(\varepsilon)$. If $m \geq 1/2$, the zero-slot fourth moment gives $1 - 2\delta m \leq 1 - \delta + O(\varepsilon)$.

No shrinking strict margin is sent into a marked moment. If $z_I \leq \nu_0$, the unmarked amplification gives $1 - \delta r \leq 1 - \delta + 2\delta\nu_0$ for $r \leq 1$. If $z_P \leq \nu_0$, the zero-slot moment gives $1 - 2\delta m \leq 1 - \delta + 6\delta\nu_0$. Both losses are within the requested output budget. The unmarked amplification is uniform through $r = 1$: its auxiliary row ball is $H = \max(2U, D^{1+c})$, $D = U^r$, with any prechosen small $c > 0$.

All subdivisions are finite or logarithmic. The ranges have $\delta \geq 1/50$ and $D_x \geq 455/222$, so their losses are uniformly bounded and can be allocated before the target. This proves (5.2). For no selected slots, use $q = 0$ in the same calculation and $t = 1$: the crossing is $r = 2/3, m = 1/3$ and yields (5.5). No positive capacity is requested in that case. $\square$

Define the balanced envelope

$$J_\kappa = (5/6 - \delta)D_x + \delta P_x, \quad t_\kappa = 1 + \frac{\delta P_x}{2J_\kappa}, \quad T_\kappa = \frac{(5/6 - \delta)\delta P_x}{2J_\kappa}, \quad R_\kappa = 1 - \delta + T_\kappa. \quad (5.8)$$

For $0 \leq \delta \leq 3/4, 0 \leq x \leq 1/2$,

$$D_x \geq 455/222, \quad P_x \geq 86/111, \quad 215/333 \leq J_\kappa \leq 5/2.$$

Indeed $D_x \geq P_x$, whence $J_\kappa \geq (5/6)P_x$ and $J_\kappa \leq (5/6)D_x \leq 5/2$. Thus $1 \leq t_\kappa < 3/2$, and substitution proves $R_{\text{short},\kappa}(t_\kappa) = L(t_\kappa) = R_\kappa$. At $\kappa = 3/4$ the sharper source bounds are $D_x \geq 37/18, P_x \geq 7/9$, and $J \geq 35/54$.

6 A general geometric criterion

Fix the box

$$\frac{1}{6} \leq \ell \leq \frac{17}{100}, \quad \frac{1}{10} \leq b \leq \frac{13}{100}, \quad (6.1)$$

and define

$$\begin{aligned} l_x &= (1 - b - \ell)/2, \quad l_y = (1 + b - \ell)/2, \quad h = (1 + b + 3\ell)/2, \quad M = 1 - \ell, \\ \sigma_g &= 11/12 - \ell/4, \quad C_g(s) = s - 2/3 - b/6. \end{aligned} \quad (6.2)$$

In (5.8) set $\kappa = 3/4$ and write J, T, R_* for its functions. Define the endpoint

$$E_g(\delta, x) = -1/4 + b/6 + 5\ell/4 - b\delta/2 - \ell(1/2 - x)\delta + hT. \quad (6.3)$$

Theorem 6.1 (Geometric criterion, including contact). *If (6.1) holds and*

$$E_g(\delta, x) \leq 0 \quad (0 \leq \delta \leq 3/4, 0 \leq x \leq 1/2), \quad (6.4)$$

then every finite-order Hecke L-function over F_0 and every Dirichlet L-function is zero-free in $\text{Re } s > \sigma_g$, with the principal pole allowed. Equality at an endpoint of (6.4) is permitted.

Proof. Geometry and common gap. Direct affine bounds on the four corners of the box give

$$\begin{aligned} l_x - \ell &\geq 9/50, \quad l_y - \ell - 11b/6 \geq 43/600, \quad 1 - 3\ell \geq 49/100, \\ 4/5 \leq h &\leq 41/50, \quad 5\ell - h \geq 11/600, \quad \sigma_g \geq 1049/1200 > 87/100. \end{aligned} \quad (6.5)$$

Thus Theorem 3.2 applies. Also

$$h = 1 - l_x + \ell, \quad C_g(s) = l_x/2 + s - 1 + h/6, \quad l_x/2 + b/12 = C_g(\sigma_g). \quad (6.6)$$

Suppose $\Delta = \beta^* - \sigma_g > 0$. This is one global number, chosen before any target. The moment parameter is $\kappa = 3/4$, with premise supplied by the input $\beta^* \leq 7/8$.

The full high identity and its contours. In the high representation from Proposition 4.2, the scalar quotient is

$$\frac{\zeta_{F_0}^S(6z)L_{F_0}^S(w, \chi_\bullet(u))}{L_{F_0}^S(s, \eta\chi_\bullet(u))},$$

with $q_u^{-z}\overline{\xi(u)}$ and the common Mellin weight

$$X^{1/2-z}Z^{s+z-1}Y^{w-1}e^{(s+z-1)^2}M_0(z)\widehat{W}_1(w).$$

Here M_0 is the scalar Mellin factor denoted M in [1, (7.2)]; it is holomorphic for $\operatorname{Re} z > 0$ and positive on $(0, 1)$. The reciprocal denominator retains its presentation and all deletions. The principal numerator row is $u = 1$ by [1, §10.1]; the other bounded unit rows remain in the small range.

For a nonprincipal original bin, first move z from 2 to $z_0 = 17/50$ with $w = 3$, then move s from 3 to $\beta^* + 20e$. All reciprocals are on global zero-free lines. Next move w to $1 - a - 6e$. Along the latter path

$$\operatorname{Re}(s + w) \geq 1 + (\beta^* - a) + 14e \geq 1 + 14e,$$

so it lies in $\mathcal{D}_1(10e)$; $6\operatorname{Re} z > 1$ and the numerator is nonprincipal. Truncate the physical heights within their assigned boxes while keeping the discarded pieces on the global s line. Only then move the retained s segment to $a + 16e$. Its reciprocal argument has real part at least $a + 6e$ and lies in the same rectangle (2.6). Hence no zero is crossed. This derives the contour move for $\sigma_g < 7/8$; it does not invoke the fixed-range conclusion of [1, Lemma 10.3].

All discarded vertical pieces and joins have

$$O_{\eta, N}(Z^{B_\eta}T_1^{-N}), \tag{6.7}$$

with a scale degree fixed before N . This is the direct input [1, Lemma 10.2]: real boxes are compact, the common kernel has pointwise rapid decay on horizontal slices as well as integrated decay, and the arithmetic integrand has a fixed scale and polynomial height bound. The latter follows from the global Hecke bounds [1, Lemmas 4.8–4.10] and Proposition 4.2. A coordinate whose reciprocal is only buffered remains in its assigned box. Internal Fourier measures with only a fixed weighted L^1 bound remain in the integrated row factor; no horizontal trace estimate is asserted for those measures. The external densities come from the original smooth profiles and have the pointwise bounds of [1, Lemma 4.5] at every fixed order.

For principal extraction keep $s_r = \beta^* + e$ and use

$$19/20 \leq w_r \leq 1 + e, \quad 33/200 \leq z_r \leq 1/6 + e.$$

These paths lie in $\mathcal{D}'_2$. First cross the numerator pole at $w = 1$, then in its residue cross $6z = 1$. There are only these two scalar poles; M_0 is holomorphic. Bound the full correction by (4.11) before approximating the double residue. Including its scale power, the combined exponent is

$$l_x/2 + s_r - 1 + hz_r + l_y(w_r - 1). \tag{6.8}$$

Thus the two leftover contours have margins

$$m_w = l_y/20 \geq 93/4000, \quad m_z = h/600 \geq 1/750, \tag{6.9}$$

with small displacement costs $(1+h)e$ and e . Take $\mu = 2\ell/(5K) > 0$, admissible in (4.10). The approximation remainder has exponent $C_g(\beta^*) + e - \mu + \varepsilon$. The double residue scalar is

$$c_S = \widehat{W}_1(1)M_0(1/6)(\text{Res}_{v=1}\zeta_{F_0}^S(v))^2/6 > 0. \quad (6.10)$$

Its sign follows from the nonnegative nonzero test and $M_0(1/6) > 0$; the finite deletion changes the positive zeta residue by $\prod_{\mathfrak{p} \in S}(1 - q_{\mathfrak{p}}^{-1}) > 0$. The factor $1/6$ is the Jacobian of $6z$. The main scale is

$$X^{1/3}Z^{s-5/6}Z^{\ell/6} = Z^{C_g(s)}.$$

Move only this main integral right to $\text{Re } s = 2$; the reciprocal is global, and Gaussian decay removes joins. Define

$$J_{\eta,g} = I_{\eta,\text{mod}}/(c_S A_T), \quad f_{\eta,g}(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{C_g(s)} e^{(s-5/6)^2} \frac{H_{\eta}(s)}{L_{F_0}^S(s, \eta)} ds. \quad (6.11)$$

Neither function involves the analysis cutoff T_1 . This derives the principal extraction for the new geometry, including its new additive constant in C_g .

Joint local errors and the retained exponent. On the central contour use the assumption-free [1, Proposition 16.1], whose local range is $51/100 \leq a \leq 1$, $0 < e \leq 10^{-3}$. Let ψ_u^* induce the numerator $\chi_{\bullet}(u)$. Its analytic hypothesis is

$$|L_{F_0}(1 - w, \overline{\psi_u^*})| \ll U^{\varepsilon} \quad (w_r = 1 - a - 6e, \quad |\text{Im } w| \leq T_1). \quad (6.12)$$

The conjugate original presentation belongs to (2.3). Its argument has real part $a + 6e$ and is in the buffered rectangle. Request $U^{\varepsilon/2}$ in (2.6), and use the inverse-deletion bound of [1, Lemma 4.10], with deletion radical $\ll_{\mathcal{A}} U$, for the remaining $U^{\varepsilon/2}$. This proves (6.12) for the primitive conjugate function. The pretarget cutoff (4.12) gives the other hypothesis, nonzero central selected $H_{\mathfrak{p}}$.

Proposition 16.1 separates each central selected multiplier into its original main factor and error. For each fixed error subset it bounds the numerator times all errors jointly by

$$U^{a-1/2+O(e)+\varepsilon}(3+T_1)^{A_{\eta}} \prod_{i \text{ merror}} P_i^{z_0-1/2+O(e)+\varepsilon}.$$

Its conductor saving is applied once to the complete set of strict ramified primes:

$$\text{Nf}_u \ll_S q_u \prod_{\mathfrak{p} \in J_0} q_{\mathfrak{p}}^{-(v_{\mathfrak{p}}(u)-1)}.$$

The main factors retain the amplitude q of Section 5. After the row count, the resulting aggregate integrand, before $q_u^{-z_0}$, is bounded by

$$U^{R+\delta/2+\varepsilon_c} Z^{\ell(z_0-1/2)+q\ell+\varepsilon_c} (1+T_1)^{A_{\eta}}. \quad (6.13)$$

There is one numerator allowance for the whole error subset.

For $e = 0$, the raw outside scale power is

$$l_x/2 + a - 1 + (1 - l_x)z_0 - al_y.$$

Add (6.13), including U^{-z_0} , and subtract $C_g(\sigma_g)$. The exact relative exponent is

$$E(d) = a - \sigma_g + h(z_0 - 1/6) - al_y - \ell/2 + q\ell + d(R + \delta/2 - z_0). \quad (6.14)$$

The adjustable costs are $(16 - 6l_y)e + (1 + d)\varepsilon_c + d\varepsilon_d + \varepsilon_p$, where ε_p pays logarithmic subdivisions. In particular the correction's $Z^{\ell z_0}$ is counted once. Bins are moved as original holomorphic integrals; pointwise amplitude subdivisions are made after this move and only used for bounds.

List once all external coordinates $v_1, \dots, v_r$ entering any buffered L -value, reciprocal, or logarithmic derivative, including the three original physical Mellin heights. After pure-twist translations write each such argument as $\gamma_j + \sum_k a_{jk}v_k$, with $|\gamma_j| \leq (3i + 1)T_1$ and one fixed finite coefficient matrix. Allocate once

$$|v_k| \leq \frac{T_1}{2r(1 + \max_j |a_{jk}|)}. \quad (6.15)$$

The total addition from all original and additional coordinates is at most $T_1/2$, and is never reset at a later transform. The original physical box constants are the corresponding entries of this same allocation. Internal seminorms are fixed before external tail orders.

All row ranges. For $R = R_*$ and $d = h$, using $a = (1 + \delta)/2$ and $q = x\delta$ in (6.14) gives exactly (6.3). Its slope in d is positive and less than two:

$$R_* + \delta/2 - z_0 = 1 - \delta/2 + T - 17/50,$$

and, since $0 \leq T \leq (5/6 - \delta)/2 \leq 5/12$,

$$33/50 - \delta/2 \geq 57/200 > 0, \quad R_* + \delta/2 - z_0 \leq 1 + 5/12 - 17/50 = 323/300 < 2.$$

The true signal has exponent $C_g(\beta^*) = C_g(\sigma_g) + \Delta$. Thus (6.4), even at equality, gives margin Δ at h .

The floor $a = 51/100$ has $R = 1$ and $q \leq 1/100$. Direct substitution gives

$$E_{\text{floor}}(h) \leq -6/25 + (32/25)\ell + b/6 \leq -11/15000 < -1/2000. \quad (6.16)$$

For $d \leq 1/2$, the no-slot bound (5.5), with the floor included, is $R = 76/75 - 2\delta/3$. This affine majorant has slope $-2/3$, equals the floor value 1 at $\delta = 1/50$, and exceeds the no-slot exponent $1 - 2\delta/3$ by $1/75$. Its slope in d is $101/150 - \delta/6 > 0$, and

$$E(1/2) \leq -2/25 - 49(1+b)/300 + 13\ell/50 + \delta(1/6 - b/4 + 3\ell/4) \leq -1631/120000 < -1/100. \quad (6.17)$$

The δ coefficient is positive, and the right side increases with ℓ and decreases with b ; its maximum is at $\delta = 3/4, \ell = 17/100, b = 1/10$.

Take $d_{\min} = 1/100$. On small rows move to $s_r = \beta^* + e, w_r = 1/2, z_r = z_0$, in $\mathcal{D}_1(1/3)$. The numerator is nonprincipal; a bounded unit denominator may be principal, whose reciprocal has no pole on this global line. Lemma 4.3 supplies the absolute tuple bound. The strip estimate [1, Lemma 4.8, (4.14)] gives numerator power $U^{3/5+\varepsilon}$, the element count is $O(U)$, and $q_u^{-z_0}$ contributes $U^{-17/50}$. Their combined row power is $63/50$. The outside scale exponent $l_x/2 + s - 1 + hz + l_y(w - 1)$ at $(s_r, w_r, z_r) = (\beta^* + e, 1/2, 17/50)$ exceeds $C_g(\beta^*) = l_x/2 + \beta^* - 1 + h/6$ by

$$h(17/50 - 1/6) - l_y/2 + e = 13h/75 - l_y/2 + e.$$

For $d \leq d_{\min}$ the row power is $(63/50)d \leq (63/50)d_{\min} \leq 2d_{\min}$. Using this larger budget, the exponent relative to $C_g(\beta^*)$ is at most

$$h(13/75) - l_y/2 + 2d_{\min} \leq (41/50)(13/75) - 93/400 + 1/50 = -2111/30000 < -7/100, \quad (6.18)$$

before arbitrarily small displacement losses.

Set

$$\lambda_0 = \min\{\Delta, 1/2000\}, \quad \zeta_0 = \min\{(5\ell - h)/4, (1 - h)/4, \lambda_0/100\} > 0. \quad (6.19)$$

Then $h + \zeta_0 < 1$, $\ell/(h + \zeta_0) > 1/5 > 7/37$, and extending the selected interval to $h + \zeta_0$ costs at most $2\zeta_0 \leq \lambda_0/50$. Its margin relative to the true signal is at least $49\lambda_0/50$.

For $q_u \geq Z^{h+\zeta_0}$ use the absolute lines $s_r = w_r = 2$, with

$$z_\infty = \lceil 10/\zeta_0 \rceil + 1 > 2.$$

Lemma 4.3 and the same outside weight give the summed exponent

$$B_0 + (h + \zeta_0)(1 + \varepsilon) - \zeta_0 z_\infty + \varepsilon < -6, \quad B_0 = l_x/2 + 1 + l_y < 2,$$

for $\varepsilon \leq 10^{-3}$. This uses a fixed absolute line before the target. Also $C_g(\sigma_g) \geq 223/1200 > 0$, so these rows have sufficient saving.

Order of choices and final height. Put $c_* = \lambda_0/100$. First request moment, witness, capacity, and rounding losses with total cost less than $c_*/10$. Obtain the target-independent fourth-moment mesh η_{mesh} . Choose a fixed even K with

$$2\ell/K < \min\{\eta_{\text{mesh}}, c_*/1000, 1/185\}, \quad \ell_i = \ell/K. \quad (6.20)$$

The bound $1/185$ ensures $1/5 - 1/185 = 36/185 > 7/37 = 35/185$ after one fractional slot is removed. Use nonnegative, nonzero bumps in the disjoint intervals

$$I_i = \left(1 + \frac{2i-1}{2K+1}, 1 + \frac{2i}{2K+1}\right).$$

At selected base U , $w_i = \ell_i/d \leq 2\ell/K$, and the supply exceeds $1/5$ by a fixed amount. The internal centered amplifier of the fourth-moment proof has pool length $\sigma_{\text{width}}/3$ and mesh below $\sigma_{\text{width}}/6$. A live physical slot is at most $2U^{\eta_{\text{mesh}}}$, whereas its auxiliary pool starts at $U^{\sigma_{\text{width}}/3}/2$. Their fixed exponent gap makes these underlying windows disjoint after a threshold, uniformly in selected d .

After fixing K , let $\mu = 2\ell/(5K)$. Choose amplitude width, e , and remaining component powers to fit the displayed finite budgets and to make principal costs less than $m_w/4, m_z/4, \mu/4$. Impose all e_0 conditions of Lemma 8.2 on the enclosing pre-saturation range $[0, 22]$. Fix $P_{\text{pre}}(e)$ at this stage. A common pre-height saving may be taken as

$$m = \frac{1}{4} \min\{c_*, m_w, m_z, \mu\} > 0.$$

All these choices, including the geometric gap, precede the target.

Now fix η , its complete presentations and S (including its defining supports and primes below P_{pre}), and all internal transform, Sobolev, and seminorm orders. The previous bounds give finite A_η, B_η and a detector ceiling $\tau_{0,\eta} > 0$, independent of an external order N . For example, if ε_{ht} is the smallest requested detector allowance and $A_{\text{ht},\eta}$ bounds its orders, take

$$\tau_{0,\eta} = d_{\min} \varepsilon_{\text{ht}} / [20(1 + A_{\text{ht},\eta})].$$

Thus for every N and every $0 < \tau \leq \min(d_{\min}/100, \tau_{0,\eta})$, after a threshold depending on η, N, τ ,

$$|J_{\eta,g} - f_{\eta,g}| \ll_{\eta,N} Z^{C_g(\beta^*)-m} (1 + T_1)^{A_\eta} + Z^{B_\eta} T_1^{-N}. \quad (6.21)$$

The external order raises only external profile derivatives; it does not reenter an internal moment. Constants in auxiliary tails can be absorbed in a threshold using a separately reserved positive

power gap. This is exactly the order required by the direct input [1, Lemma 11.1], whose boundary range is $(1/2, 1)$.

Choose

$$0 < \tau_\eta \leq \min\{d_{\min}/100, \tau_{0,\eta}, m/[4(1 + A_\eta)]\},$$

then choose N_η so that $B_\eta - N_\eta \tau_\eta < C_g(\beta^*) - m/2$, and finally take the threshold. Equation (6.21) gives a saving $m/2$. Theorem 3.2 and $A_T^{-1} = Z^{o(1)}$ give $|J_{\eta,g}| \ll Z^{C_g(\sigma_g) + \Delta/2}$. Lemma 4.1 gives a holomorphic H_η with $|H_\eta - 1| \leq 1/31$ on $\text{Re } s > \sigma_g$. Proposition 2.1 applies with common losses $\Delta/2, m/2$ and contradicts $\Delta > 0$. The transfer in Section 2 completes the theorem. $\square$

6.1 Coefficients for the endpoint certificates

For general $c = 1/(3\kappa)$, put $y = 1/2 - x$ and use J_κ, T_κ from (5.8) in (6.3). Direct multiplication gives

$$-108J_\kappa E_g = Q_0(\delta) + yQ_1(\delta) + y^2Q_2(\delta) + y^3Q_3(\delta),$$

where $Q_0 = A_0\delta^2 + B_0\delta + C_0$, $Q_1 = A_1\delta^2 + B_1\delta + C_1$, $Q_2 = A_2\delta^2 + B_2\delta$ and $Q_3 = 216c\ell\delta^2$. Write $e_0 = -1/4 + b/6 + 5\ell/4$ and $h = (1 + b + 3\ell)/2$. The coefficients are

$$\begin{aligned} A_0 &= 27bc/2 - 54b - 81c\ell/2 - 27c/2 + 81\ell + 27, \\ B_0 &= -54(c - 3)e_0 + 45b(5 - 2c)/2 - 45h(2 - c)/2, \\ C_0 &= -45(5 - 2c)e_0, \\ A_1 &= -108bc + 108b + 54c\ell + 54, \\ B_1 &= 126bc - 18b + 180c\ell - 54c - 45\ell - 18, \\ C_1 &= -30bc - 15b - 225c\ell + 45c - 225\ell/2 + 45/2, \\ A_2 &= 162bc - 54c\ell + 54c + 108\ell, \\ B_2 &= -81bc - 225c\ell + 9c + 90\ell. \end{aligned}$$

These follow by expanding $-108J_\kappa\{e_0 - (b/2 + \ell y)\delta\} - 54h(5/6 - \delta)\delta P_x$. At $c = 4/9$ they give

$$\begin{aligned} A_0 &= -48b + 63\ell + 21, & B_0 &= 98b + 120\ell - 52, \\ C_0 &= -185b/6 - 925\ell/4 + 185/4, \\ A_1 &= 60b + 24\ell + 54, & B_1 &= 38b + 35\ell - 42, \\ C_1 &= -85b/3 - 425\ell/2 + 85/2, \\ A_2 &= 72b + 84\ell + 24, & B_2 &= -36b - 10\ell + 4, & Q_3 &= 96\ell\delta^2. \end{aligned}$$

7 Two rational stages

Theorem 7.1 (First rational improvement). *Every function in Theorem 1.1 is zero-free in*

$$\text{Re } s > \frac{349999}{400000} = \frac{7}{8} - \frac{1}{400000}.$$

Proof. Set $\eta_0 = 1/100000$ and

$$\ell = \frac{1}{6} + \eta_0, \quad b = \frac{1}{8}, \quad l_x = \frac{17}{48} - \frac{\eta_0}{2}, \quad l_y = \frac{23}{48} - \frac{\eta_0}{2}, \quad h = \frac{13}{16} + \frac{3\eta_0}{2}.$$

The geometry belongs to (6.1) and $\sigma_g = 349999/400000$. The exact positive margins are

$$l_x - \ell = \frac{37497}{200000}, \quad l_y - \ell - \frac{11b}{6} = \frac{49991}{600000}, \quad 1 - 3\ell = \frac{49997}{100000}, \quad 5\ell - h = \frac{12521}{600000}.$$

We reproduce the pure algebra of [1, Lemma 20.2], using it only for its original numerical function. Write E_0 for (6.3) at $\ell = 1/6, b = 1/8, y = 1/2 - x, v = 51 + 41y$, and

$$p_y = 7 + 18y + 8y^2, \quad j_y = 185 + 170y + (-138 + 12y + 96y^2)\delta.$$

Then $D_x = (37 + 34y)/18$, $P_x = p_y/9$, $J = j_y/108$, and the explicit identities are

$$10368J(-E_0) = 10(37 + 34y) - 8(237 + 377y + 26y^2)\delta + 48(51 + 131y + 94y^2 + 32y^3)\delta^2, \quad (7.1)$$

$$10368vJ(-E_0) = (3 + 5y)\{(4v\delta - 79)^2 + 49\} + 4y\{4v\delta[(1 + 3y)(15 + 32y)\delta + 9 - 13y] + 265 + 3485y\}. \quad (7.2)$$

For $0 \leq \delta \leq 3/4$, $0 \leq y \leq 1/2$, every term on the last line is nonnegative, since $9 - 13y \geq 5/2$. Moreover $v \leq 17(3 + 5y)$ and $J \leq 5/2$. Consequently $E_0 \leq -49/440640$. The exact increment for our new geometry is

$$E_g - E_0 = \eta_0\{5/4 - y\delta + 3T/2\} \leq 15\eta_0/8,$$

since $0 \leq T \leq (5/6 - \delta)/2 \leq 5/12$. Thus

$$E_g \leq -\frac{49}{440640} + \frac{15}{800000} = -\frac{20369}{220320000} < -\frac{9}{100000}.$$

Theorem 6.1 applies. Its proof already supplies all nonendpoint ranges and contours for this geometry. $\square$

Theorem 7.2 (Second rational improvement, independently obtained by Gao [2]). *Every function in Theorem 1.1 is zero-free in*

$$\operatorname{Re} s > \frac{20999}{24000} = \frac{7}{8} - \frac{1}{24000}.$$

Proof. Take

$$\ell = \frac{1001}{6000}, \quad b = \frac{1}{8}, \quad l_x = \frac{4249}{12000}, \quad l_y = \frac{5749}{12000}, \quad h = \frac{3251}{4000}.$$

These lie in (6.1) and give the stated σ_g . Let $y = 1/2 - x$. Here

$$D_x = (37 + 34y)/18, \quad P_x = (7 + 18y + 8y^2)/9, \quad 35/54 \leq J \leq 5/2.$$

Multiplication of the complete endpoint by its positive denominator gives the polynomial identity

$$\mathcal{P} := 24000(-108JE_g) = Q_0(\delta) + yQ_1(\delta) + y^2Q_2(\delta) + y^3Q_3(\delta), \quad (7.3)$$

where

$$\begin{aligned} Q_0 &= 612252\delta^2 - 473520\delta + 91575, \\ Q_1 &= 1572096\delta^2 - 753860\delta + 84150, \\ Q_2 &= 1128336\delta^2 - 52040\delta, \quad Q_3 = 384384\delta^2. \end{aligned}$$

The general coefficients in Section 6.1 give this identity at $c = 4/9$; `anc/verify_rational.py` also checks it with exact rational arithmetic. Equivalently, expand

$$-108J\{-1/4 + b/6 + 5\ell/4 - (b/2 + \ell y)\delta\} - 54h(5/6 - \delta)\delta P_x.$$

The discriminant of Q_0 is -46717200 and its vertex is $19730/51021 > 1/3$. Its minimum is $324425/17007 > 19$. The exact completed-square remainders

$$4(1572096)(84150 + 6500) - 753860^2 = 1737110000 > 0,$$

$$4(1128336)(1000) - 52040^2 = 1805182400 > 0$$

prove $Q_1 \geq -6500$ and $Q_2 \geq -1000$ on the whole real axis.

Cover the closed rectangle by three intervals in δ . If $0 \leq \delta \leq 3/10$, Q_0 is decreasing and

$$\mathcal{P} \geq Q_0(3/10) - 6500/2 - 1000/4 = 28042/25 > 19.$$

If $3/10 \leq \delta \leq 1/3$, then $Q_0 \geq Q_0(1/3) = 1763$, Q_1 is increasing and $Q_1(3/10) = -12984/25$, while $Q_2, Q_3 \geq 0$. Thus

$$\mathcal{P} \geq 1763 - 6492/25 = 37583/25 > 19.$$

For $\delta \geq 1/3$, put $v_0 = \delta - 1/3$. The exact expansions

$$Q_1 = 1572096v_0^2 + 294204v_0 + 22622/3,$$

$$Q_2 = 1128336v_0^2 + 700184v_0 + 108024,$$

$$Q_3 = 384384v_0^2 + 256256v_0 + 128128/3$$

are nonnegative, so $\mathcal{P} \geq Q_0 > 19$. Since $J \leq 5/2$,

$$E_g \leq -\frac{19}{6480000} < -\frac{1}{350000} < 0$$

on the entire rectangle. Theorem 6.1 proves this stage without using the first rational theorem. $\square$

8 The quadratic endpoint of the original envelope

Theorem 8.1. *Every function in Theorem 1.1 is zero-free in*

$$\operatorname{Re} s > \sigma_* := \frac{1507 - 2\sqrt{921}}{1653}.$$

Proof. Put $r_0 = \sqrt{921}$ and

$$d_* = \frac{49 - r_0}{48}, \quad \ell_* = \frac{33 + 8r_0}{1653}, \quad b_* = -\frac{4}{29} + \frac{230r_0}{26709}. \quad (8.1)$$

We first derive these parameters. At $y = 1/2 - x = 0$, the numerator $N = -108JE_g$ is

$$Q_0(\delta) = (-48b + 63\ell + 21)\delta^2 + (98b + 120\ell - 52)\delta - 185b/6 - 925\ell/4 + 185/4.$$

The coefficient of b vanishes when

$$288\delta^2 - 588\delta + 185 = 0.$$

The root in $[0, 3/4]$ is d_* . Substitution in (5.8) gives $R_*(d_*, 1/2) = 2/3$, equivalently $T = d_* - 1/3$. Setting the endpoint to zero yields

$$\ell_* = \frac{5 - 6d_*}{9 + 18d_*}.$$

The stationary equation $Q'_0(d_*) = 0$ then gives

$$b_* = \frac{52 - (126d_* + 120)\ell_* - 42d_*}{98 - 96d_*},$$

which reduces to (8.1). Also $\sigma_* = 11/12 - \ell_*/4$.

The rational isolation

$$\frac{30347}{1000} < r_0 < \frac{30348}{1000} \quad (8.2)$$

follows from

$$921 - (30347/1000)^2 = 59591/10^6 > 0, \quad (30348/1000)^2 - 921 = 1104/10^6 > 0.$$

It implies $1/6 < \ell_* < 17/100$, $1/10 < b_* < 13/100$, and $d_* > 777/2000$.

For the complete rectangle numerator, set

$$\begin{aligned} A_0 &= -48b_* + 63\ell_* + 21, & B_1 &= 38b_* + 35\ell_* - 42, \\ A_1 &= 60b_* + 24\ell_* + 54, & B_2 &= -36b_* - 10\ell_* + 4. \\ C_1 &= -85b_*/3 - 425\ell_*/2 + 85/2, \\ A_2 &= 72b_* + 84\ell_* + 24, \end{aligned}$$

Expansion gives the identity

$$N = A_0(\delta - d_*)^2 + y(A_1\delta^2 + B_1\delta + C_1) + y^2(A_2\delta^2 + B_2\delta) + 96\ell_*y^3\delta^2. \quad (8.3)$$

The double root in the constant coefficient is a consequence of the two displayed elimination equations, not a numerical approximation.

All sign comparisons are in $\mathbb{Q}(r_0)$. Reduce a field expression to $A + Br_0$ using $r_0^2 = 921$, and use the appropriate endpoint of (8.2). This gives

$$A_0 > 25, \quad A_1 > 65, \quad A_2 > 46.$$

For the closest comparisons, the exact field expressions are

$$4A_1(C_1 + 6/25) - B_1^2 = \frac{17317418985607 - 570341476744r_0}{6990413025} > 1, \quad (8.4)$$

$$4A_2(3/125) - B_2^2 = \frac{-6767947591748 + 223064581728r_0}{34952065125} > 1/30, \quad (8.5)$$

$$Q'_1(3/10) = -2729/145 + 117028r_0/133545 > 7,$$

$$Q_1(3/10) + 1/200 = 2140057/66120 - 40590578r_0/38060325 > 0,$$

$$Q_1(1/3) = 52541/1653 - 1578082r_0/1522413 > 3/10,$$

$$(3/10)A_2 + B_2 = 1956/145 - 6736r_0/133545 > 11.$$

Here $Q_1 = A_1\delta^2 + B_1\delta + C_1$, $Q_2 = A_2\delta^2 + B_2\delta$, and $Q_3 = 96\ell_*\delta^2$. Equations (8.4)–(8.5) give $Q_1 \geq -6/25$, $Q_2 \geq -3/125$ by completing squares.

If $0 \leq \delta \leq 3/10$, then

$$Q_0 > 25(777/2000 - 3/10)^2 > 19/100, \quad N > 19/100 - 3/25 - 3/500 = 8/125 > 0.$$

If $3/10 \leq \delta \leq 1/3$, then

$$Q_0 > 25(777/2000 - 1/3)^2 > 3/40.$$

The derivative guard makes Q_1 increasing there, so $Q_1 > -1/200$, and the Q_2 guard makes $Q_2 \geq 0$. Thus $N > 3/40 - 1/400 = 29/400 > 0$. If $1/3 \leq \delta \leq 3/4$, then $Q_1 > 0$ and $Q_2, Q_3 \geq 0$. Hence $N \geq 0$, with equality only at $y = 0, \delta = d_*$. Since $J > 0$, this proves $E_g \leq 0$ on the entire closed rectangle. Theorem 6.1 includes the contact point and proves the claimed half-plane.

For comparison,

$$\frac{7}{8} - \sigma_* = \frac{16r_0 - 485}{13224} > 0, \quad \ell_* - \frac{1001}{6000} > \frac{1}{3306000} > 0.$$

The second strict bound follows by evaluating at the lower endpoint in (8.2) and using the strict root inequality. Thus this stage improves both rational stages. $\square$

9 Extending the plain fourth moment

Theorem 9.1 (Extended fourth moment). *Retain the arithmetic and coefficient class of [1, Lemma 18.1] as follows. Rows are $0 < q_k \ll Z^m$ and $\psi_k(n) = \tau(n)\chi_n(k)$, with τ fixed within this row sum. Let q_{union} bound the logarithmic norm of the complete union of its displayed moving supports, before cancellation, and put $M = m + q_{\text{union}}$. Any additional squarefree polynomial-norm coprimality mask is fixed within the row sum and acts simultaneously on both plain factors and every slot. All natural zeros remain.*

Let Θ be a fixed finite ray-character group containing the fixed twists, reciprocity phases, and $n \mapsto \chi_n(-1)$. Each positive-length slot has a coefficient which is a fixed finite linear combination of members of Θ , independent of Z and the row. The underlying prime windows are disjoint before masks. Let $\mathcal{R}_0$ consist of rows with nonprincipal primitive inducing character and, for $z > 0$, let $\mathcal{R}_z$ consist of rows whose inducing character is outside Θ .

For prescribed bounded nonnegative length ranges, every $\varepsilon > 0$, and

$$37/50 \leq \kappa \leq 1,$$

there is a mesh $\eta_{\text{mesh}} > 0$, uniform in this interval and chosen before the slot count and arithmetic datum, such that, if $z_i \leq \eta_{\text{mesh}}$ and $z = \sum_i z_i > 0$,

$$\sum_{k \in \mathcal{R}_z} |S_{\psi_k}(n_1; W_1) S_{\psi_k}(n_2; W_2) Q_k|^2 \ll Z^{M+\varepsilon} \quad (9.1)$$

provided

$$n_1 + n_2 + 6\kappa z \leq M, \quad \kappa < 1 \implies \beta^* \leq (1 + \kappa)/2. \quad (9.2)$$

For a fixed system and datum, finite seminorm and height orders may depend on them but are uniform in all moving labels and admissible masks. The zero-slot conclusion for $\mathcal{R}_0$ holds for arbitrary bounded nonnegative n_1, n_2 , without a zero-free premise.

Proof. We reconstruct the finite induction of [1, §18], replacing all inequalities that used its lower bound $\kappa \geq 3/4$. Its zero-slot case at $\kappa = 1$ is already available without the Part II standing supposition, as proved in Appendix A. All positive-slot recursive calls below refer to the new assertion at smaller effective width.

Masks, reflection, and the prime bound. Let $T_k(X)$ be a natural centrally normalized plain sum. For an additional fixed mask R , the exact identities [1, (18.4)–(18.6)] are

$$S_{k,R}(X) = \sum_{d|R} \mu(d) \psi_k^0(d) q_d^{-1/2} T_k(X/q_d),$$

$$Q_{k,i,R} = Q_{k,i}^0 - P_i^{-1/2} \sum_{\substack{\mathfrak{p}|R \\ \mathfrak{p} \text{ prime}}} \psi_k^0(\mathfrak{p}) \nu_i(\mathfrak{p}) W_i^{\text{slot}}(q_{\mathfrak{p}}/P_i).$$

Apply these simultaneously to both plains and all slots. The complete divisor mass costs an arbitrary small power; extracted scalars retain their zeros. This operation only decreases $n_1 + n_2 + 6\kappa z$ for $\kappa > 0$.

The primitive functional equation and restoration of the redundant radical give the reflection scale bound of [1, (18.7)–(18.9)]:

$$C_k q_{R_{0,k}} \ll_{\mathcal{A}} Z^M, \quad Y = C_k q_{d_0} / (X q_{h_0}) \ll_{\mathcal{A}} Z^M / X, \quad d_0 \mid R_{0,k}, \text{ rad}(h_0) \mid R_{0,k}.$$

The product of conductor and redundant radical is bounded by the complete row and moving-support union. The gamma quotient maps the original annular test to a test bounded at zero and rapidly decreasing at infinity, with fixed finite seminorm and polynomial height costs. Partition it into whole annuli, discard upper tails at $Z^{\xi/2}$ with fixed orders, and clip retained subunit scales using their fixed support endpoints. After a threshold each retained reflected length satisfies

$$n_{\text{ref}} \leq M - n + \xi. \quad (9.3)$$

Reflection first gives the conjugate character; conjugating that whole factor inside its absolute value restores the common character and conjugates its profile. Natural zeros are unchanged. No reciprocal contour is used in this reflection.

Set $s_\kappa = (1 + \kappa)/2 \in [87/100, 1]$. When $\kappa < 1$, (9.2) allows the global logarithmic-derivative bound of [1, Lemma 4.9] on $s_\kappa + e$. For each coefficient character $\vartheta \in \Theta$, the inducing character of $\psi_k \vartheta$ is nonprincipal, since otherwise ψ_k would induce a member of Θ . The smoothed von Mangoldt Mellin shift has no pole. Removing prime powers and the original finite deletions by [1, Lemma 4.10], with conductor and radicals in their polynomial ranges, gives for the whole slot product

$$|Q_k|^2 \ll Z^{\kappa z + \varepsilon_1} p_j(\mathbf{W})^b (1 + |\omega|)^h. \quad (9.4)$$

The exponent is $2s_\kappa - 1 = \kappa$, and the total displacement cost is $2ez$. Derivatives and heights are not charged in that exponent. For $\kappa = 1$, absolute prime counting gives the same estimate. All real losses are uniform on $[37/50, 1]$, since the logarithmic bound allows the entire s_κ range, not just $s_\kappa \geq 7/8$.

Terminal widths and same-width comparison. Write $A = n_1 + n_2 + z$. The new interval gives the exact inequalities

$$86/25 \leq 6\kappa - 1 \leq 5, \quad z \leq 25M/111 < M/4, \quad \kappa z \leq M/6. \quad (9.5)$$

The original terminal bound $z \leq 2M/9$ is replaced by $25M/111$; its actual squared cost remains at most $M/6$ by (9.4). At width $M \leq \rho$, the positive-slot case follows from the completed zero-slot case and (9.4): since $A \geq z$, (9.2) gives $6\kappa z \leq M$, so the extra exponent is $\kappa z \leq M/6 \leq \rho/6$, the term $\rho/6$ in T_{term} below.

If $A > 5M/6$, then $(6\kappa - 1)z \leq M - A < M/6$, so

$$z < 25M/516 < M/20.$$

Put $L = M/4$. The equal-product comparison has lengths $L, A - z - L$, both at least L . If an original plain length is below L , reflect the other one. Otherwise introduce the comparison with the same two profiles, and reflect its longer factor only when bounding that comparison. Equation (9.3) gives

$$A_{\text{comp}} \leq 3M/2 - A + 2z + \xi \leq 23M/30 + \xi, \quad (9.6)$$

$$A_{\text{comp}} + (6\kappa - 1)z \leq 5M/2 - 2A + 2z + \xi \leq 14M/15 + \xi. \quad (9.7)$$

Each boundary has at least $M/15$ margin before ξ . For $M > \rho$ take $\xi \leq \rho/30$. Thus this comparison belongs to the already completed uncentered part of its width band.

The common coefficient and genuine transforms. For allocated plain powers $\mathbf{b} = (b_1, b_2)$ preserve

$$D_{\mathbf{b}}(l_1, l_2) = \prod_{i=1}^2 W_i(q_{b_i} q_{l_i} / X_i) - \prod_{i=1}^2 W_i(q_{b_i} q_{l_i} / Y_i), \quad X_1 X_2 = Y_1 Y_2,$$

or its first rectangle for an uncentered input. The coefficient on the full product $u = \prod l_1 l_2$ is

$$\left(\prod_i \nu_i(\mathfrak{p}_i) W_i^{\text{slot}}(q_{\mathfrak{p}_i} / P_i) \right) D_{\mathbf{b}}(l_1, l_2) \tau_1(u) q_u^{it} \mathbf{1}_{(u, R)=1} \mathbf{1}_{s|u}.$$

It is exactly the class of [1, (18.17)–(18.18)]: slots may share primes with either plain, and the mask, character, norm power, and allocated powers are common to both rectangles. No zero scalar is removed by division.

In the uncentered region, majorize the nonnegative squared norm by the full smooth row ball. In the centered region, first bound the reflected comparison on $\mathcal{R}_z$ as above; then majorize the nonnegative squared norm of the entire difference by the full smooth row ball. The following Poisson formulas apply only to those full balls, never to an indicator selecting inducing characters.

Let $\theta_{\text{slots}} = H_{\text{slots}} / \log Z$ bound all fixed support-ratio errors for every subset, as in [1, (18.19)]. One fixed H_{slots} is selected after the datum and slot count. Whole-dyad tail removal is performed on genuine sums before an off-coprime extension. The retained full kernel, inverse roots, normalized row norm, and two whole-product norms are then separated by one common Fourier measure. The source finite identities [1, Lemmas 13.2–13.4, (18.21), (18.30)] apply to these exact full characters and supports. CRT is used on the genuine coprime locus; its extension is multiplied by the complete finite Möbius indicator before independent factors are bounded. None of these identities contains κ .

Here are the normalization ledgers, with the local log-norm labels of [1, (18.22)–(18.44)]. Let c, d be the full common-support powers in the first pair, p their common radical, R_1 the net nonprincipal support, E_1 the complementary row-mask divisor, and $s_0 = \log_Z q_s$ the full mutual-gcd Möbius label. Put $\tilde{q} = q_{\text{union}} + R_1 + E_1$ and

$$K_0 = 2A - c - d + R_1 + E_1 - m, \quad K \leq K_0 + \delta_{\text{fr},1}, \quad 0 \leq \delta_{\text{fr},1} < \xi.$$

The first Poisson zero term is $O(Z^{m+\varepsilon_1})$. On a nonzero side it suffices to bound

$$\mathcal{N}_C \ll Z^{A-c+\tilde{q}+B_c-s_0+\varepsilon_G}, \quad B_c = (3c - 5d - R_1)_+ / 6$$

and the corresponding expression $B_d = (3d - 5c - R_1)_+ / 6$. Including the outside factor, common-support count, and both sides of Cauchy, the exact exponent is

$$\begin{aligned} & (m - A - R_1/2 - E_1) + (p + s_0) \\ & + \frac{1}{2} \{ A - c + \tilde{q} + B_c - s_0 + A - d + \tilde{q} + B_d - s_0 \} \\ & = M + \frac{1}{2} \{ B_c + B_d - (c + d - 2p - R_1) \} \leq M. \end{aligned} \tag{9.8}$$

For completeness, the last inequality is the primewise integer inequality in [1, (18.24)]. At a common prime with multiplicities $i \geq j \geq 1$, let $r = \mathbf{1}_{6 \nmid i-j}$. The positive contribution is at most $(3i - 5j - r)_+ / 6$ and the available length is $i + j - 2 - r \geq 0$. When the former is positive, six times the difference is $3i + 11j - 12 - 5r \geq 0$: for $r = 0$ it is at least 2; for $r = 1$, $i \geq j + 1$ gives at least

0. The positive part of a sum is bounded by the sum of positive parts. Summing with the prime log-norm weights proves (9.8).

For Gauss-row enlargement choose width step $\sigma_{\text{width}} > 0$, pool length $\ell_p^* = \sigma_{\text{width}}/3$, and mesh below $\sigma_{\text{width}}/6$. The pool is disjoint from all live windows after a threshold. Enlarge each Gauss norm once. The three extracted valuations are 1, 6, 7. If $\ell_p = \log_Z q_p \in [\sigma_{\text{width}}/6, \sigma_{\text{width}}/3]$, their removal and new-support pairs are

$$(w, w_o) = (\ell_p, \ell_p), (6\ell_p, 0), (7\ell_p, \ell_p), \quad 0 \leq w_o \leq w \leq 7\sigma_{\text{width}}/3.$$

The squared extraction coefficients are respectively q_p^{-1} , $(1 - q_p^{-1})^2$, q_p^{-1} . The row phases and punctures multiply both rectangles. Errors are not amplified again. The original averaging loses one pool density, and a fixed output row has boundedly many representations; there is no extra prime-count factor.

Put

$$a_0 = A - c - w, \quad J_0 = d - c + (K_0 - K) - 2w + w_o.$$

The smooth ball uses $g = (J_0)_+ + 2\sigma_{\text{width}}$ and $\ell_{\text{stage}} = \sigma_{\text{width}}/3$ for the main amplifier; errors and zero-slot norms use $g = (J_0)_+ + \sigma_{\text{width}}$ and $\ell_{\text{stage}} = 0$. Add the row zero only at this final ball. The local [1, (13.5)] makes $G(u, 0)$ vanish unless u is a sixth power. With $s \mid u$, its squared contribution is at most

$$Z^{a_0/3+4\theta_{\text{slots}}/3+\varepsilon_1} \leq Z^{a_0-s_0+2\theta_{\text{slots}}+\varepsilon_1},$$

since $a_0 \geq -\theta_{\text{slots}}$ and $s_0 \leq (a_0 + \theta_{\text{slots}})/6$. Zero is never multiplied by a pool prime.

In the second complete extraction let c_2, d_2 be the common-support powers, $b_2 = (c_2 + d_2)/2$, p_2 their radical, g_2 the smaller multiplicity sum, t_2 its unit/nonprincipal radical contribution, and V_2 its nonunit radical. Its nominal child widths are

$$\begin{aligned} m' &= 2a_0 - K - g - g_2 - V_2, & q' &= \tilde{q} + w_o + t_2 + V_2, \\ M' &= m' + q' = M + J_0 - g - g_2 + t_2 \leq M - \sigma_{\text{width}}. \end{aligned} \tag{9.9}$$

This uses $g - J_0 \geq \sigma_{\text{width}}$ and $g_2 \geq t_2$. The outside exponent is

$$K + g - \ell_{\text{stage}} - a_0 + p_2 - s_0 + g_2 - t_2 - b_2.$$

Subtracting it from the first-side allowance $a_0 + \tilde{q} + w_o + w + B_c - s_0$ leaves precisely

$$M' + \Delta_{\text{child}}, \quad \Delta_{\text{child}} = b_2 - p_2 + w + B_c + \ell_{\text{stage}} \geq w + \ell_{\text{stage}}. \tag{9.10}$$

The radical count is $Z^{p_2-s_0+\varepsilon_1}$, because s divides that radical. The second enclosing width satisfies $0 \leq M'_{\text{act}} - M' < \delta_{\text{fr},2} < \xi$. All local correlation bounds are applied on their exact unit/nonunit partitions, extended by zero off the retained support.

Actual child normalization and whole-slot removal. The coefficient result [1, Lemma 18.2] applies: both transforms, common-support extraction, and the full Möbius allocation preserve one character, one mask, one norm power, and equal-product rectangles on each separated side. Quotients after a divisor extraction are not repunctured at that divisor. The two sides' characters differ by a member of Θ .

We display the clipping calculation to preserve the actual normalization. For one side and rectangle let x_i be the pre-extraction formal plain lengths, $a_i \geq 0$ the extracted divisor lengths, and $y_i = x_i - a_i$. Let e_j be the total frozen-slot ratio error, ω_j the newly frozen-slot ratio error, and $e_{r,j}$ the sum of its two divisor lengths and frozen slot lengths. Retained nonempty profiles have

$$\pi_{0,j} = \sum_i (-x_i)_+, \quad \pi_j = \sum_i (-y_i)_+, \quad 0 \leq \pi_{0,j} \leq \pi_j \leq \theta_{\text{slots}}, \quad r_{\text{clip},j} = e_{r,j} - (\pi_j - \pi_{0,j}) \geq 0.$$

Writing $\alpha_j = a_0 - c_2$ or $a_0 - d_2$, their exact clipped total is

$$A_{\text{clip},j} = \alpha_j + e_j + \pi_{0,j} - r_{\text{clip},j}.$$

A divisor dyad of lower length t^- has $e_{r,j} + \omega_j \geq t^-$ and $|e_j + \omega_j| \leq \theta_{\text{slots}}$. Thus its paired count and extraction factors satisfy

$$t^- - \frac{r_{\text{clip},1} + r_{\text{clip},2}}{2} + \frac{e_1 + \pi_{0,1} + e_2 + \pi_{0,2}}{2} \leq 2\theta_{\text{slots}}.$$

Nonexceptional centered children are first bounded by their two rectangles and then legally clipped. Empty rectangles are tested by fixed support endpoints; retained arithmetic and profile zeros stay. Exceptional rectangles remain formal and are not clipped.

For a nonexceptional actual child put

$$F_{\text{act}} = (A_{\text{act}} - M'_{\text{act}} + (6\kappa - 1)z_{\text{act}})_+.$$

The parent inequality, $z_{\text{act}} \leq z$, and [1, (18.38)] give

$$F_{\text{act}} \leq 6(w + \ell_{\text{stage}}) + \delta_{\text{fr},1} + 2\theta_{\text{slots}}. \quad (9.11)$$

To verify the only nontrivial stage comparison, before clipping

$$(A' - M') - (A - M) = g + w - d - c_2 - (K_0 - K) - w_o + g_2 - t_2.$$

Use $g_2 - t_2 \leq c_2$. For the main amplifier the remainder is at most $6\ell_{\text{stage}} + \delta_{\text{fr},1}$. For an error it is at most $w - w_o + \sigma_{\text{width}} + \delta_{\text{fr},1}$, and the three displayed valuation pairs give $w - w_o + \sigma_{\text{width}} \leq 6w$. The clipped and enclosed corrections add at most $2\theta_{\text{slots}}$. This proves (9.11) for the whole new κ interval, using only $6\kappa - 1 \geq 0$.

If necessary, remove whole slots with (9.4). A slot of length d_z lowers the constraint by $6\kappa d_z$ and costs κd_z in the squared exponent. The first greedy prefix reaching $F_{\text{act}}/(6\kappa)$ overshoots by at most one mesh slot, so

$$\begin{aligned} d_z &\leq \min\{z_{\text{act}}, F_{\text{act}}/(6\kappa) + \eta_{\text{mesh}}\}, \\ \kappa d_z &\leq F_{\text{act}}/6 + \eta_{\text{mesh}} \leq \Delta_{\text{child}} + \delta_{\text{fr},1}/6 + \theta_{\text{slots}}/3 + \eta_{\text{mesh}}. \end{aligned} \quad (9.12)$$

The ratio is exactly $1/6$, independent of κ . If all slots disappear before the constraint is met, use the already completed unrestricted zero-slot assertion. Otherwise the remaining product satisfies the new positive-slot condition at smaller width. Cauchy averages the two sides' costs and does not double them. The per-edge loss beyond the child envelope is at most

$$\delta_{\text{fr},2} + \delta_{\text{fr},1}/6 + \eta_{\text{mesh}} + 7\theta_{\text{slots}}/3. \quad (9.13)$$

Exceptional rows and the diagonal. For a child inducing in Θ , fix its unit and S -valuation sectors. Let v_1 be the radical length of the nonunit primes counted in V_2 whose two common multiplicities are one, and put $f = 2v_1$. The retained common-support zeros exclude an old moving character at these primes. The fixed factor of the new row has valuation two, so an exceptional inducing character forces the varying row valuation to be four modulo six. At every other good prime its valuation is likewise fixed modulo six by the frozen character. Thus $(h') = h_0 a^6$, with h_0 fixed and sixth-power-free, and $q_{h_0} \geq Z^{2f}$. Ideal counting gives the stronger exponent $(m'_{\text{act}} - 2f)/6$ on a nonempty ball. We use only the weaker count from [1, (18.41)]:

$$Z^{(m' - f)/6 + O(\xi) + \varepsilon_1},$$

including rows added after a partition scalar is bounded on a positive norm. No extra forcing saving is assumed from its unit-prime partition. Define

$$\begin{aligned} F_1 &= c/6 + 5\{d + (K_0 - K)\}/6 + w/3 + \tilde{q}/6 + w_o + B_c - 5g/6 + \ell_{\text{stage}}, \\ F_2 &= 2b_2 - 5g_2/6 - p_2 + t_2 + V_2/6 + f/6. \end{aligned}$$

The exact remaining normalization identity is

$$(m' - f)/6 + a_0 - b_2 - (M' + \Delta_{\text{child}}) = A - 5M/6 - F_1 - F_2.$$

Its lower bounds are

$$F_1 \geq 2(c + w)/3 - 3\sigma_{\text{width}} - 5\delta_{\text{fr},1}/6, \quad F_2 \geq 2b_2/3.$$

For the first bound, temporarily set $g = (J_0)_+$ and $\ell_{\text{stage}} = 0$, and put $D_0 = d + K_0 - K$. If $J_0 \geq 0$, substitution gives $F_1 = c + 2w + \tilde{q}/6 + w_o/6 + B_c \geq c + 2w$. If $J_0 < 0$, use $\tilde{q} \geq R_1$ and $D_0 \geq d - \delta_{\text{fr},1}$ to obtain

$$B_c + \tilde{q}/6 \geq (3c - 5D_0)_+/6 - 5\delta_{\text{fr},1}/6.$$

Hence $F_1 \geq 2(c + w)/3 - w/3 - 5\delta_{\text{fr},1}/6$. Restoring $g \leq (J_0)_+ + 2\sigma_{\text{width}}$ costs at most $5\sigma_{\text{width}}/3$, and $w/3 \leq 7\sigma_{\text{width}}/9$. Their sum is $22\sigma_{\text{width}}/9 < 3\sigma_{\text{width}}$.

For the second bound, the primewise contributions, in units of the prime log-norm, are

common case	F_2	b_2
$i = i, 6 \nmid i, \text{ unit}$	$7i/6$	i
$i = i, 6 \nmid i, \text{ nonunit}$	$(7i - 5)/6 + \frac{1}{3}\mathbf{1}_{i=1}$	i
$i = i, 6 \mid i, \text{ either}$	$7i/6 - 1$	i
$i > j_0, 6 \mid j_0, \text{ unit}$	$i + j_0/6 - 1$	$(i + j_0)/2$

The difference from $2b_2/3$ is nonnegative: in the second row it is zero at $i = 1$ and $(3i - 5)/6$ for $i \geq 2$; in the third it is $(i - 2)/2$ with $i \geq 6$; in the last it is $(4i - j_0 - 6)/6$ with $j_0 \geq 6, i \geq j_0 + 1$. Summing proves the bound. Thus for $A \leq 5M/6$ the uncentered excess costs only $3\sigma_{\text{width}} + O(\xi)$.

For centered exceptional children, Lemma 18.2 leaves the same fixed character, mask, and norm power in both rectangles. Use the assumption-free [1, Lemma 18.3] when all four formal scales have a positive lower length; otherwise use its absolute-volume fallback at every positive scale, including subunit scales. Its main coefficient is independent of scale and norm twist, so equal products cancel exactly. The gain from [1, (18.48)–(18.49)] is $(L - v)_+$, where $v = c + w + \min(c_2, d_2)$ and $L = M/4$. The remaining deficit is bounded by

$$A - 5M/6 - 2v/3 - (L - v)_+ \leq A - M \leq 0.$$

For $v \leq L$ its maximum occurs at L , and for $v \geq L$ it again occurs at L . The parent condition gives $A \leq M$, because $6\kappa - 1 > 0$. The second Poisson diagonal of [1, (18.31)] uses this same inequality and costs at most $5\sigma_{\text{width}}/3 + O(\xi)$. Neither argument uses the former lower endpoint $3/4$.

Finite depth and choices. Complete zero-slot widths first. Then process positive-slot width bands in increasing order, proving the uncentered $A \leq 5M/6$ assertion before the centered remainder. The comparison (9.6)–(9.7) invokes only that already completed uncentered region at the same width. True recursive calls decrease actual width by at least $\sigma_{\text{width}}/2$ after frequency enclosures. If M_{max} bounds initial widths, take

$$D_{\text{depth}} = 2 + \lceil 2M_{\text{max}}/\sigma_{\text{width}} \rceil.$$

A branch has at most $D_{\text{depth}} - 2$ strict calls.

Let $C_* \geq 1$ bound the finitely many aggregate affine coefficients and positive-part Lipschitz constants in the displayed stage ledgers, including the local errors in [1, (18.42)–(18.50)]. This bound is independent of slot count: its aggregate list is

$$A, z, c, d, p, R_1, E_1, s_0, K, w, w_o, g, \ell_{\text{stage}}, c_2, d_2, p_2, g_2, t_2, V_2.$$

Uniformity on the new interval follows from $0 \leq 6\kappa - 1 \leq 5$ and the ratio $1/6$ in (9.12). This derives the propagation constant anew rather than applying the old moment theorem outside its range.

For $0 < \varepsilon \leq 1$, choose

$$\rho = \varepsilon/100, \quad \sigma_{\text{width}} = \varepsilon/1000, \quad \delta_{\text{pad}} = \varepsilon^2/10^6.$$

The terminal reserve is

$$T_{\text{term}} = \rho + \delta_{\text{pad}} + \rho/6 + 3\sigma_{\text{width}} + 5\sigma_{\text{width}}/3 < \varepsilon/4.$$

Before slot count or datum, take

$$\begin{aligned} \xi &< \min\{\delta_{\text{pad}}/4, \rho/60, \sigma_{\text{width}}/(8C_*), \varepsilon/(32C_*D_{\text{depth}})\}, \\ \eta_{\text{mesh}} &< \min\{\sigma_{\text{width}}/6, \varepsilon/(32C_*D_{\text{depth}})\}, \\ \varepsilon_0 &< \varepsilon/(32C_*D_{\text{depth}}). \end{aligned}$$

These choices also preserve the comparison margin. Use the envelope $T_{\text{term}} + (d+1)C_*(\eta_{\text{mesh}} + \xi + \varepsilon_0)$ at depth d . Each branch pays one terminal reserve; triangle takes a maximum and Cauchy averages. The total loss is less than ε .

Now fix the slot count, datum, and profiles, form the finite-depth support boxes, and take Z large enough that the single θ_{slots} and all fixed support factors fit the tolerances. Whole raw tails were removed by [1, Lemma 4.7] on the genuine sums; fixed weighted Fourier moments propagate by [1, Lemma 17.3 and Corollary 17.4]. Choose internal prime, reflection, and exceptional-mask orders backwards through the finite graph. They are finite and introduce no factor $Z^{J\xi}$. Any later external tail order differentiates only external profiles. Finally restore the masks by simultaneous deletion, using the preallocated arbitrary small power. This proves (9.1); losses larger than one follow by using a smaller loss. $\square$

10 Self-consistency and the cubic endpoint

Theorem 10.1 (Self-consistent geometric criterion). *Use (6.1)–(6.2), and set*

$$\kappa_b = 2\sigma_g - 1 = 5/6 - \ell/2, \quad c_b = 1/(3\kappa_b).$$

Define E_b by (6.3), replacing T with T_{κ_b} from (5.8). If $E_b \leq 0$ on $[0, 3/4] \times [0, 1/2]$, then the entire Hecke and Dirichlet families are zero-free in $\text{Re } s > \sigma_g$.

Proof. The box gives $449/600 \leq \kappa_b \leq 3/4$. Suppose $\Delta = \beta^* - \sigma_g > 0$. The actual parameter used in Theorem 9.1 is

$$\kappa_a = 2\beta^* - 1 = \kappa_b + 2\Delta \in (\kappa_b, 3/4]. \quad (10.1)$$

Its zero-free premise holds by equality. The baseline κ_b is only an algebraic comparison parameter; no moment is invoked with it under this supposition.

Choose the branch and t by the baseline crossing (5.6). On the inverse branch the count is unchanged. On the plain branch the plain-witness length satisfies $m \geq 1/3 - O(\varepsilon)$, and replacing the baseline capacity by the actual admissible capacity increases the exponent by

$$\begin{aligned} 2q(1-2m) \left(\frac{1}{6\kappa_b} - \frac{1}{6\kappa_a} \right) &= \frac{(2q)(1-2m)\Delta}{3\kappa_b\kappa_a} \\ &\leq \frac{\Delta}{9\kappa_b\kappa_a} + O(\varepsilon) < \Delta/4 + O(\varepsilon). \end{aligned} \quad (10.2)$$

Here $2q \leq 1$ and $9(37/50)^2 = 12321/2500 > 4$. Near zero capacities use the unselected fallback from Theorem 5.1; all losses remain arbitrary small powers. Thus the actual count is bounded by $R_{\kappa_b} + \Delta/4 + \varepsilon$.

At $d = h$, the exponent relative to the true signal is at most

$$E_b + h\Delta/4 - \Delta \leq -(1 - h/4)\Delta \leq -159\Delta/200. \quad (10.3)$$

The floor, intermediate, small, and principal estimates of Theorem 6.1 are unchanged, since they were proved on the entire geometry box. Take

$$\lambda_0 = \min\{159\Delta/200, 1/2000\}, \quad 0 < \zeta_0 < \min\{(5\ell - h)/4, (1 - h)/4, \lambda_0/100\}.$$

The actual slope in the selected range remains less than two, including (10.2); indeed $\Delta \leq 1/1200$, $T \leq 5/12$, and $\delta \leq 3/4$ give $R_{\kappa_b} + \Delta/4 + \delta/2 - z_0 \leq 323/300 + 1/4800 < 2$. Its extension costs at most $2\zeta_0$. Use the uniform mesh of Theorem 9.1, then the even slot count, disjoint bumps, remaining real losses, e , and pretarget cutoff exactly in the order proved in Section 6. Only afterwards choose the target, internal orders, height, external order, and threshold. This gives positive common high saving, low loss $\Delta/2$, and the same nonvanishing H_η and physical J_η . Proposition 2.1 and the transfer prove the assertion. $\square$

Theorem 10.2 (Cubic stage). *Let $\sigma_\dagger$ be as in Theorem 1.1. Every function there is zero-free in $\text{Re } s > \sigma_\dagger$.*

Proof. Define

$$F(u) = 1188u^3 - 2160u^2 + 171u + 190 \quad (10.4)$$

Its values modulo 7 at $u = 0, \dots, 6$ are 1, 5, 3, 4, 3, 2, 3. Since 7 does not divide 1188 and F has no root modulo 7, this cubic is irreducible modulo 7, hence its primitive polynomial is irreducible over $\mathbb{Q}$, and the exact calculations below take place in the cubic field it defines. Let $d_\dagger$ be its root in

$$I = \left[\frac{3885833}{10^7}, \frac{3885834}{10^7} \right]. \quad (10.5)$$

This root is in the interior and is unique there, since

$$\begin{aligned} F(3885833/10^7) &= \frac{13153091803447489}{25000000000000000000} > 0, \\ F(3885834/10^7) &= -\frac{1385634440463739}{31250000000000000000} < 0, \end{aligned}$$

and on $(19/50, 2/5)$

$$F'(u) = 3564u^2 - 4320u + 171 \leq 3564(2/5)^2 - 4320(19/50) + 171 = -22509/25 < -900.$$

Put $d = d_{\dagger}$ and define

$$\begin{aligned} \ell_{\dagger} &= \frac{5-6d}{9+18d}, & \sigma_{\dagger} &= \frac{7+18d}{9+18d}, & \kappa_b &= \frac{5+18d}{9+18d}, & c &= \frac{3+6d}{5+18d}, \\ b_{\dagger} &= -\frac{8(216d^2+18d-13)}{3(2d+1)(792d^2-618d-203)}. \end{aligned} \quad (10.6)$$

All denominators are nonzero on I ; in particular $-324 < 792u^2 - 618u - 203 < -323$ and $-176 < 288u^2 - 318u - 95 < -175$ there. These formulas give $\sigma_{\dagger} = 11/12 - \ell_{\dagger}/4$ and $\kappa_b = 2\sigma_{\dagger} - 1$.

Here is the derivation. At $x = 1/2$, forcing $R_{\kappa}(d, 1/2) = 2/3$ removes the endpoint's coefficient of b . Setting that endpoint to zero gives $\ell = (5-6d)/(9+18d)$. Substitution of the self-consistency $\kappa = 5/6 - \ell/2$ gives the rational identity

$$R_{\kappa}(u, 1/2) - 2/3 = -\frac{F(u)}{6(288u^2 - 318u - 95)}. \quad (10.7)$$

It can be checked without rational-function division: with $J_{\text{num}} = (5/6 - u)(19 + 78u) + u(7 + 30u)$, the polynomial identity is

$$2J_{\text{num}}(1/3 - u) + (5/6 - u)u(7 + 30u) = F(u)/18.$$

With $c = (3 + 6u)/(5 + 18u)$ and $x = 1/2$, the definitions give

$$J_{\kappa} = \frac{J_{\text{num}}}{2(5 + 18u)}, \quad J_{\text{num}} = -\frac{288u^2 - 318u - 95}{6}.$$

Consequently,

$$R_{\kappa}(u, 1/2) - 2/3 = \frac{2J_{\text{num}}(1/3 - u) + (5/6 - u)u(7 + 30u)}{2J_{\text{num}}} = \frac{F(u)}{36J_{\text{num}}}.$$

Finally, at fixed ℓ, c , the stationary equation for the endpoint numerator is

$$(4752d^3 - 1332d^2 - 3072d - 609)b + 1728d^2 + 144d - 104 = 0.$$

The left side is $-(4/3)(2d+1)(18d+5)$ times $Q'_0(d)$, with $\ell = \ell_{\dagger}$ and $c = (3+6d)/(5+18d)$ held fixed in the derivative. Since its coefficient of b is $3(2d+1)(792d^2-618d-203)$, its solution is precisely $b_{\dagger}$.

We now certify the full rectangle. Let $y = 1/2 - x$ and $N = -108J_{\kappa_b}E_b$. With $\ell = \ell_{\dagger}$, $b = b_{\dagger}$,

$$N = Q_0(\delta) + yQ_1(\delta) + y^2Q_2(\delta) + y^3Q_3(\delta), \quad Q_0 = A_0(\delta - d)^2, \quad (10.8)$$

where Q_1, Q_2, Q_3 and A_0 are the general coefficients of Section 6.1 evaluated at these ℓ, b, c . Equations (10.7) and the two parameter equations give $Q_0(d) = Q'_0(d) = 0$, proving its squared form exactly in $\mathbb{Q}[d]/(F(d))$.

The following are explicit rational interval certificates. In the table, the entry (L, U) means $L/10^6 < v < U/10^6$ (weak inequalities at these enlarged endpoints would also suffice). They follow by direct interval operations on the displayed rational functions over I : add endpoints, take the four endpoint products for multiplication, and take the reciprocal of a denominator interval excluding zero. No reduction by F is needed for these bounds. The intervals for ℓ, b, c are computed directly from I before rounding; the table rows are then rounded outward to 10^{-6} . Using the rounded ℓ, b, c rows in the coefficient formulas gives wider intervals that still imply every inequality used below.

Using the upper root endpoint and s_+ gives

$$\sigma_* - \sigma_{\dagger} > \frac{27350173}{543750000000000} > \frac{1}{20000000}.$$

□

11 The envelope limit and one further mean-value hypothesis

11.1 The scope of the endpoints

The following concerns only the optimisation of (ℓ, b) inside the fixed set of estimates listed here; it says nothing about the true value of β^* or about other methods. The endpoints just proved concern a specified envelope: $M + \ell = 1$, the low exponent $l_x/2 + b/12$, the signal C_g , the marked inverse and plain moments, the unmarked sixth-power amplification, and an absolute high comparison on every allowed bin. They impose no lower bound on actual row contributions.

For the original $\kappa = 3/4$ envelope, the critical bin of Section 8 satisfies

$$E_g(d_*, 1/2) = (3/4 + 3d_*/2)(\ell - \ell_*),$$

independently of b . Thus requiring every raw endpoint to be nonpositive forces $\ell \leq \ell_*$; Theorem 8.1 attains that bound.

For the extended, self-consistent envelope the same obstruction occurs at $\sigma_{\dagger}$. Here is the precise local range of this assertion. If $8749/10000 \leq \beta \leq \sigma_{\dagger}$, use the smallest legal plain-moment parameter $\kappa = 2\beta - 1$ and let $c = 1/(3\kappa)$. At $x = 1/2$, for $0 \leq u < 5/6$, $R_c(u)$ is strictly decreasing and crosses $2/3$ at one $d_c \in (19/50, 2/5)$. Indeed, with $v = P_x/D_x$,

$$R_c(u) = 1 - u + \frac{u}{2} \frac{v(5/6 - u)}{5/6 - u + uv}, \quad \frac{d}{du} \frac{v(5/6 - u)}{5/6 - u + uv} = -\frac{(5/6)v^2}{(5/6 - u + uv)^2} < 0,$$

so $R'_c(u) < -1/2$. The endpoint signs are

$$R_c(19/50) - 2/3 = \frac{587c - 698}{150(193c - 454)} > 0, \quad R_c(2/5) - 2/3 = \frac{c + 11}{15(38c - 89)} < 0$$

for $4/9 \leq c \leq 5000/11247$. Increasing κ cannot improve this count, since

$$\partial_c R_c(u) = -\frac{u(6u - 5)^2}{2(6cu - 10c - 18u + 25)^2} \leq 0.$$

Put $\ell_\beta = 4(11/12 - \beta)$ and $d_\beta = (9\beta - 7)/(18(1 - \beta))$. The map $u \mapsto (7 + 18u)/(9 + 18u)$ increases. Equation (10.7), together with $F' < 0$, gives $d_c \geq d_\beta$, hence $\ell_c = (5 - 6d_c)/(9 + 18d_c) \leq \ell_\beta$. At this legal critical bin, the high exponent relative to the true signal is

$$(3/4 + 3d_c/2)(\ell - \ell_c) - (\beta - \sigma_g).$$

Strict low saving requires $\ell > \ell_\beta$. At $\ell = \ell_\beta$ this expression is nonnegative, and its derivative in ℓ is $1/2 + 3d_c/2 > 0$. It therefore cannot give a strict high saving. This proves the stated envelope limitation, including geometry allowed to depend on β . A different mean value, correlation, signed cancellation, or low-side estimate lies outside this model.

11.2 A single unproved estimate

The family and central normalization are those of [8, (2.2), Proposition 3.1], which proves the mean-square bound on the nonzero row ball for $H = D^{1+\vartheta}$, $0 < \vartheta \leq 1/10$; the hypothesis below asks for its smooth-majorant form for $D^{99/100} \leq H \leq D^{101/100}$. Fix a finite-order ray character ν with its entire zero-extended presentation, a finite S containing its defining-modulus support and the primes over 6, a compact annular interval $I = [a, b] \subset (0, \infty)$, and a fixed nonnegative radial Schwartz majorant $\Phi(q_u/H)$ with $\Phi \geq 1$ on $[0, 1]$ and compactly supported radial Fourier transform. Use the self-dual Eisenstein lattice measure $d\mu(z) = (2/\sqrt{3}) dx dy$. Thus $\hat{\Phi}(0) = \int_{\mathbb{C}} \Phi(|z|^2) d\mu(z)$. Define

$$\begin{aligned} A_u(D) &= \sum_{(n,S)=1} \mu(n) \nu(n) \chi_n(u) W(q_n/D), \\ \mathcal{M}(D, H) &= \frac{1}{D} \sum_{u \in \mathcal{O}} \Phi(q_u/H) |A_u(D)|^2. \end{aligned} \quad (11.1)$$

Hypothesis 11.1 (Mean-square estimate for $H \geq D^{99/100}$; unproved). For every fixed datum ν, S, I, Φ just specified and every $\varepsilon > 0$, there are a finite integer J and a constant $C_{\nu,S,I,\Phi,\varepsilon}$ such that

$$\mathcal{M}(D, H) \leq C_{\nu,S,I,\Phi,\varepsilon} H D^\varepsilon \|W\|_{C^J(I)}^2 \quad (11.2)$$

for all $D \geq 2$, $D^{99/100} \leq H \leq D^{101/100}$, and $W \in C_c^\infty(I)$, with every original natural zero retained. The exponent 99/100 is common to all fixed data; J and C may depend on the displayed datum and ε .

This is one mean-square estimate. It is not proved in this paper and is not an input to Theorem 1.1.

Proposition 11.2 (Conditional 328/375 boundary). *Hypothesis 11.1 implies that all the functions of Theorem 1.1 are zero-free in $\operatorname{Re} s > 328/375$.*

Proof. By (11.2), $\mathcal{M} \ll H D^\varepsilon \|W\|_{C^J}^2$. Since $\Phi \geq 1$ on the unit row ball, this bounds the unweighted ball when $H \geq D^{99/100}$. For larger H the existing unmarked bound [1, (17.87)] covers the range, with a small fixed positive length loss. Apply the same assumption to the finitely many scale derivatives and then the profile Sobolev estimate of [1, Lemma 4.5].

For completeness, sixth-power-free amplification changes its long-row threshold as follows. Put $M_u(D; W) = D^{-1/2} A_u(D; W)$. The original zero extensions give the exact divisor identity

$$M_u(D; W) = \sum_{d \mid \operatorname{rad} a} \mu(d) \psi_u(d) q_d^{-1/2} M_{ua^6}(D/q_d; W), \quad \psi_u(n) = \nu(n) \chi_n(u).$$

To check it, split each original squarefree column uniquely as $n = dl$ with $d \mid \operatorname{rad} a$ and $(l, a) = 1$. Multiplicativity retains every zero, including when u, a share primes; the factor $q_d^{-1/2}$ is its exact central normalization. Average this identity over good primary ideals a of norm at most $P = (H'/U)^{1/6}$, where $H' = \max(2U, D^{99/100})$, and apply weighted Cauchy and the scale supremum just established. The map $(u, a) \mapsto ua^6$ is injective when u is sixth-power-free. The good-ideal count is $\asymp_{\mathcal{A}} P$ (the bounded-ratio case is handled directly), so the resulting bound is H'/P times arbitrary small powers. Divisor multiplicities cost arbitrary small powers and no coprimality of u, a is imposed. The normalized unmarked moment therefore has exponent

$$e_{99}(r) = \max\{1, 1/6 + 33r/40\} \quad (D = U^r),$$

up to the requested loss and a fixed height cost. The long count for $r \geq 1$ is

$$\max\{1 - \delta, 1/6 + (33/40 - \delta)t\} + O(\varepsilon).$$

This includes the new kink at $r = 100/99$.

Use $\kappa = 3/4$ for the marked counts. Set

$$J_{99} = (33/40 - \delta)D_x + \delta P_x, \quad t_{99} = 1 + \frac{D_x/120 + \delta P_x/2}{J_{99}}, \quad T_{99} = \frac{(97/120 - \delta)\delta P_x}{2J_{99}}.$$

Then $1 \leq t_{99} < 3/2$, since $97/120 > 3/4$, and substitution balances the short count with the second long branch at $R_{99} = 1 - \delta + T_{99} \geq 1 - \delta$. The same inverse margins, supply, zero-capacity, and coefficient arguments of Theorem 5.1 apply.

Choose

$$\ell = 21/125, \quad b = 11/100, \quad l_x = 361/1000, \quad l_y = 471/1000, \quad h = 807/1000.$$

This is in (6.1) and gives $\sigma_g = 328/375$. Replace T in (6.3) by T_{99} to define E_{99} . For $y = 1/2 - x$, the full polynomial identity is

$$\begin{aligned} 20000(-108J_{99}E_{99}) &= 526080\delta^2 - 406288\delta + 79365 \\ &\quad + y(1292640\delta^2 - 603308\delta + 72930) \\ &\quad + y^2(920640\delta^2 - 19144\delta) + 322560y^3\delta^2. \end{aligned}$$

The three exact completed-square remainders are

$$\begin{aligned} 4(526080)(79365 - 900) - 406288^2 &= 45529856 > 0, \\ 4(1292640)(72930) - 603308^2 &= 13108397936 > 0, \\ 4(920640)(100) - 19144^2 &= 1763264 > 0. \end{aligned}$$

Thus the four coefficients have lower bounds 900, 0, -100, 0, respectively, on $\delta \geq 0$. For $0 \leq y \leq 1/2$ the numerator is at least 875. Since $77/120 \leq J_{99} \leq 99/40$,

$$E_{99} \leq -7/42768 < 0.$$

All other ranges, contours, normalizer, and choice order were proved uniformly on the geometry box. The count just derived replaces the selected envelope there; its slope is positive and less than two. The common-signal contradiction and transfer therefore prove the conditional boundary. No second unproved estimate is used. $\square$

A Auxiliary results independent of the Part II supposition

The first paragraph of [1, Part II, p.85] supposes $\beta^* > 7/8$ and defines its dynamic parameter $\kappa = 2\beta^* - 1$. Our uses of its auxiliary results require versions with this supposition removed. The following proposition specifies those versions and proves the removal by their mathematical dependencies.

Proposition A.1. *The local statements in the following list remain valid with their original coefficient classes, zero extensions, bounded parameter ranges, masks, and uniformity, without the standing supposition $\beta^* > 7/8$:*

Lemmas 13.1–13.4; Corollary 14.1; Lemmas 14.2–14.3;

Proposition 15.2; Proposition 16.1;

Lemmas 17.1–17.3; Corollary 17.4; Lemmas 17.5–17.6;

Lemma 18.1 in its printed range, Lemmas 18.2–18.3; Lemma 19.1.

For positive slots, Lemma 18.1 retains its explicit premise $\beta^ \leq (1 + \kappa)/2$ when $\kappa < 1$. Proposition 16.1 retains the explicit primitive conjugate-numerator bound (6.12). Lemma 19.1 retains its buffered and cumulative-frequency hypotheses. All other results in this list need no assertion about the location of Hecke zeros.*

Proof. Finite correlations and marked reflection. Lemma 13.1 is the prime ideal asymptotic in a fixed ray quotient, with a fixed annular test and fixed finite deletions. Its proof uses the prime ideal theorem in a fixed Frobenius class [1, §13.2] and Stieltjes integration by parts; neither the quotient nor the deletion set moves with the prime scale. Its proof does not use the family supremum. Lemma 13.2 evaluates prime-power Fourier sums. Lemma 13.3 is finite Fourier inversion of two complete Gauss sums, retaining their common factors and the original principal zero convention. Lemma 13.4 is its CRT factorization after the complete shared prime support is extracted; the residual factors are genuinely coprime there. These are finite identities. They apply to the whole products and exact masks retained in Sections 3 and 9.

Corollary 14.1 applies the Part I completed cubic reflection, [1, Proposition 5.1], to the product-form whole-index mark. Its conditions are a fixed finite ray family, disjoint fixed prime lists, coefficients independent of the row and fourth-power label, and the original puncture. Expand each local $1 - \chi_p^0$. The inactive coefficient difference is q_p^{-1} , an active mark has exponent zero, and the row exponent is one. CRT gives the row-slot cubic exponent $1 + 0 + 2 = 3$, whose phase is one. The remaining two zero-preserving symbols are those of [1, (14.16)]. This reasoning contains no reciprocal of a Hecke function and no geometry.

Lemma 14.2 uses the squarefree indices k, n, P , positive norm scales $K, N, B, L \geq 1$, and two coefficient-independence conditions: $a(P)$ is independent of k, n, b and $\beta(n, b)$ is independent of k, P . The indices k, P avoid S ; the source n, b may contain primes of S other than λ . Expand the full (P, k) mask, write $P = DP'$, $k = Dk'$, and freeze the other divisor and row labels before applying the quadratic reduction [1, (5.27)] and the cubic norm [1, (14.6)]. In the cubic step the squarefree column $j = c'm = n/r$ may contain primes of S other than λ ; [1, p.96] passes it to [1, (14.6)] after cubic reciprocity and conjugation, since (14.6) excludes no primes. Its output is

$$(K + NB)B\{N + L + (NL)^{2/3}\},$$

up to small powers. The residual divisor factors only improve this expression. There is no zero-location premise.

Lemma 14.3 combines these two facts with the common kernel and lattice tails of [1, Lemmas 5.3–5.4]. It is applied on actual residual-row dyads, with bounded lengths, fixed characters and punctures, disjoint product-form slots, and row restrictions independent of active slots and dual variables except for their explicit coprimality mask. The exact joint kernel is [1, (14.15)], including every dual and slot norm. Its squared scalar saving is $(T_d - v - 3\ell_b - e_\lambda)_+/2$. The hybrid norm gives (3.7); summing the fixed powerful and supported row parts adds $O/2$ once. The kernel contours are

those of the completed theta reflection, not reciprocal Hecke contours. Thus neither the standing supposition nor the geometry [1, (12.4)] enters this proof.

The additive norm and local errors. Proposition 15.2 assumes fixed $S, T, b_{\text{aux}}, \xi, W_1, c_T$, polynomial $Q, Y' \geq 1$, $P_a = Y'^2/Q \geq 1$, a fixed row ball, and precisely its finite ray-coefficient class. Expanding its square and applying Lemma 13.3 gives the full common-factor correlation. Freeze the fixed ray parts before periodicizing the joint coefficient. On nonexceptional frequencies a nontrivial good-prime character has zero complete mean. Four-dimensional lattice Poisson gives the bound in [1, (15.5)], and summing the exact divisor and row counts yields $(Q/Y')(P_a^2/Y')$. Exceptional frequencies have $(k) = a_1^6 e$, with e supported on the extracted factor and fixed set; elementary counting gives $(Q/Y')P_a^{1/6}$. The zero frequency is the original diagonal and contributes Q/Y' . These are exactly the three terms in [1, (15.4)]. No use is made of the geometry-dependent Proposition 15.3.

Proposition 16.1 assumes its displayed real strip, sixth-power-free physical rows outside S , nonprincipal primitive numerator, nonzero central selected local factors, and (6.12). Its proof uses the six-case table (4.1) and one conductor allocation to the entire set of strict ramified primes. The functional-equation conductor exponent is $a - 1/2 + 6e$; its validity requires $51/100 \leq a \leq 1$, not $a > 7/8$. Section 6 establishes its explicit conjugate premise by the common buffer. Lemma 4.3 supplies its cutoff without a target-dependent choice. The separate principal-factor paragraph [1, (16.12)–(16.14)] is excluded; Proposition 4.2 proves its replacement on $\mathcal{D}'_2$.

Finite propagation and masked Poisson. Lemma 17.3 and Corollary 17.4 are finite backward propagation on an acyclic graph. Their hypotheses are fixed real terminal and edge exponents, finite weighted moments of one common coefficient measure, fixed-dimensional smooth profile bounds, and fixed linear height maps. Normalizing by the prescribed norm exponent makes every label measure have bounded small-power mass. Backward induction raises finite seminorm and height orders but leaves the real norm powers fixed. All these statements concern nonnegative integrals and finite graph depth; no zero-location assertion occurs.

Lemma 17.5 is masked primitive Poisson. With a primitive finite character ψ modulo $\mathfrak{m}$, an ideal R , a radial Schwartz test, and $K > 0$, its formula is

$$\sum_k \psi(k) \mathbf{1}_{(k,R)=1} \Phi(q_k/K) = \frac{K\gamma(\psi; \mathfrak{m})}{\sqrt{q_{\mathfrak{m}}}} \sum_{d \mid \text{rad } R} \frac{\mu(d)\psi(d)}{qd} \sum_h \overline{\psi(h)} \mathcal{F}\Phi\left(\frac{Kq_h}{qdq_{\mathfrak{m}}}\right).$$

Expand the complete mask, substitute $k = dk'$, and use finite Fourier inversion modulo $\mathfrak{m}$. The finite transform is $\sqrt{q_{\mathfrak{m}}}\gamma(\psi; \mathfrak{m})\overline{\psi(h)}$. Primitivity kills nonunit frequencies; Parseval gives $|\gamma| = 1$ for a nonprincipal primitive character. For the primitive principal modulus use $\mathfrak{m} = 1$, $\gamma = 1$, and value one at zero. At each use here the row scale is at least one. Its principal nonzero-frequency restoration costs $O(KZ^\varepsilon)$ by lattice counting and Schwartz decay, with the complete outer mask retained. This is a finite-arithmetic and lattice statement.

Canonical marked estimate and its initialization. For precision the canonical family of Lemma 17.2 has fixed puncture $\rho(n) = \mathbf{1}_{(n,r_\rho)=1}$, fixed finite-ray ν , disjoint product-form prime mark $d(n)$ of total length at most z_0 , and squarefree labels $q_f \asymp Z^V$ with a nonnegative divisor-bounded weight $w(f)$ depending only on f . Put $E = N + V$. Its two local premises are

$$q_{r_\rho} \leq Z^{E-M-z_0-c_*}, \quad 3M + 6z_0 + c_* \leq 4E, \quad c_* > 0.$$

The normalized family is

$$Z^{-V} \sum_f w(f) \sum_{0 < q_k \ll Z^M} \left| Z^{-N/2} \sum_{n \text{ sf}} \overline{\alpha(n)} \gamma_2(n) \nu(n) \rho(n) \chi_n(k) \chi_n(f)^4 d(n) W(q_n/Z^N) \right|^2.$$

No additional arbitrary column coefficient is allowed.

Its terminal proof uses the exact cube-Möbius completion [1, (17.10)] and the assumption-free Lemma 14.3. The node invariants are

$$E - M - Q - z_0 \geq c_{\text{node}}, \quad 4E - 3M - 6z_0 \geq c_{\text{node}}, \quad Q = \log_Z q_{r_\rho}.$$

For step $d \leq c_*/200$, $c_{\text{node}} \geq c_*/2$, and support and local losses at most $c_*/1000$, the three reflected branches [1, (17.13)–(17.14)] are strictly below E :

$$E - c_{\text{node}} + 5\eta/3, \quad E - 2c_{\text{node}} + 5d + 17\eta/3 + \tau_{\text{ref}}, \quad E - M/4 - 3c_{\text{node}}/4 + 5d/2 + 19\eta/6 + \tau_{\text{ref}}/2.$$

The complete weight in the f -sum cancels its outside Z^{-V} ; no extra label count is inserted into the hybrid norm.

The remaining canonical terms are reduced by the two finite masked Poisson transforms [1, (17.15)–(17.65)]. Priority slot allocation is an exact partition even at shared primes. The first transform is used on genuine coprime columns, its raw tail is removed before full mutual-gcd Möbius extension, and its kernel is one joint profile with all norm roots retained. The old f label is removed once by the divisor-fiber count [1, (17.32)]. The second transform reconstructs a canonical squarefree label $f_{\text{new}} = JCd_2v'$; its squarefreeness follows from the retained coefficient zeros, not a new mask. Its puncture is fixed within that child, independent of its varying row and label. The child margins in [1, (17.51)] lose at most 5η and 7η , and its row width drops at least $2d - 8\eta$. The exact normalization [1, (17.64)] is

$$\kappa_i + \lambda_c + C_c + 2F_c = E,$$

where κ_i is a normalization exponent, not a moment parameter. With $\eta \leq d/16$, the per-edge loss is $40\eta + \tau + \pi$. Choose the finite depth and then η, τ, π so that the sum of these losses is below the requested ε , as in [1, (17.66)]. All raw tail orders are chosen after the crude count, and the finite propagation rule fixes internal orders before an external cutoff. These invariants and transformations contain neither β^* nor a dynamic-bin inequality.

Lemma 17.1 follows by its single masked-Poisson initialization [1, (17.67)–(17.86)]. The exact overlap split is $n = jn_0$, $P = jP_0$; the scalar Z^{-G} pays the assigned prime count. The resulting canonical label is squarefree and its full puncture remains within its local premise. For prescribed inverse margins $c_1, c_2 > 0$, the child margins are at least $c_1 - 9\eta - \tau_{\text{init}}$ and $c_2 - 22\eta - 3\tau_{\text{init}}$, so $c_* = \min(c_1, c_2)/2$ is allowed after small losses. The energy on each side is $m + 11\eta + \pi_{\text{init}} + \varepsilon_c$. Cauchy averages rather than doubles these costs. This derives the marked inverse estimate under its stated strict margins with no family-boundary input.

Finally [1, (17.87)–(17.90)] use its zero-slot specialization, scale Sobolev, and the exact zero-preserving divisor identity. On sixth-power-free rows the map $(u, a) \mapsto ua^6$ is injective. This proves Lemma 17.6 under its printed $H = \max(2U, D^{1+c})$ and bounded profile ranges, again without a zero-location hypothesis.

The fourth moment and prime amplitudes. For Lemma 18.1, simultaneous mask deletion and primitive reflection [1, (18.4)–(18.10)] use only the general smooth calculus, functional equation, and deleted-factor estimates. Its sole zero-location input is [1, (18.11)] on $s_\kappa = (1 + \kappa)/2 \geq \beta^*$. It is absent for zero slots and replaced by absolute prime counting at $\kappa = 1$. The terminal widths, same-band comparison, two complete finite transforms, paired clipping, and exceptional cancellation [1, (18.13)–(18.50)] use only the local length inequalities and κ . The finite-depth choices [1, (18.51)–(18.52)] close the estimates. Section 9 rederives every inequality affected by extending that parameter below $3/4$.

Lemma 18.2 is preservation of the complete common coefficient under those transforms and full Möbius allocations; Lemma 18.3 is lattice Poisson with one fixed character, one common polynomial-norm squarefree mask, one norm power per side, and the four original formal scales. Their equal-product leading terms cancel, and their absolute-volume fallback is valid at every positive scale. These results have no zero-location premise.

Lemma 19.1 is a buffered prime-annulus consequence of Part I Lemma 8.2 and Lemma 4.9. The prime-annulus frequency is within the same cumulative allowance, and the main slot retains its original identity-ray restriction and zero extension. Its logarithmic derivative is controlled only on that buffer; its exterior tails stay on the absolute line. Thus its only analytic premise is the stated buffer, not the standing contradiction.

This establishes each item of the list with exactly the retained local hypotheses. $\square$

The fixed-geometry conclusion of [1, Proposition 15.3], the positive-gap capacity replacement in [1, Proposition 19.2], and [1, Proposition 20.3] are not inputs here. Nor are the fixed-boundary conclusions of [1, Lemmas 10.3–10.6] or [1, Lemma 16.2] used outside their printed ranges. Their needed parameter-dependent estimates were derived in Sections 3–10. The pure identity in [1, Lemma 20.2] was reproduced in (7.1)–(7.2).

The paragraph *The principal local factor* of [1, §16.2], namely [1, (16.12)–(16.14)], is not an input either: it is stated on the second Euler region [1, (7.15)] ($\text{Re } s \geq 7/8$) and is replaced by Proposition 4.2 on $\mathcal{D}'_2$.

B Printed inputs and their hypotheses

The tables specify the statements used with their printed parameters. A local version means the version proved in Proposition A.1, with every local coefficient restriction retained. Proof identities from the input are used only with their displayed original coefficients and supports. The new geometry, Euler domain, completed norm, row envelope, and fourth-moment parameter are proved in the body.

Input in [1]	Hypotheses retained	Application and verification
Theorem 1.1; (2.1)	The whole primitive finite-order Hecke family over F_0 ; poles excluded from β^* .	Gives $\beta^* \leq 7/8$ and the exact ceiling $a \leq \beta^*$ for actual-zero bins.
Proposition 2.1	$1/2 < \sigma < 1$, $C(s) = s + c$, positive global gap, target-independent low and high losses, one J, f , bounded nonvanishing H on $\text{Re } s > \sigma$.	Proposition 2.1 states and proves this input. Section 6 verifies its hypotheses, including algebraic endpoint equality.
§6.1; (6.1)–(6.2)	Fixed ray group T ; complete zero-extended presentations supported on fixed S ; nonnegative nonzero annular W_0, W_1 ; arbitrary positive scales; $Q = q_{b_{\text{aux}}}XY$; original masks.	The scales are positive powers of Z . Retaining the whole-index indicator linearly in the absolute series derives the marked separation in Theorem 3.2.
Lemma 4.5	Fixed logarithmic boxes; one common coefficient measure for the row Hilbert space; specified finite weighted moments; uniformly smooth profiles. Sobolev parameters enter profiles, not arithmetic coefficients.	The low proof keeps row, dual, and slot norms in one profile. Witness choices are smooth-parameter choices. Equations (6.15) and (6.21) separate external and internal orders.

Input in [1]	Hypotheses retained	Application and verification
Lemma 4.6	Fixed positive truncation width, Gaussian annular seminorms summable with exponential weights, fixed polynomial-scale range.	The joint Gaussian profile meets these conditions. Half-width truncation and a threshold control whole retained annuli.
Lemma 4.7	Schwartz Fourier kernel, fixed normalized radial derivatives; bounded real Mellin strips and fixed weighted derivatives of the tests.	Genuine raw tails are removed before off-coprime extension. Raising orders changes seminorms, not real exponents.
Lemma 4.8; (4.14)	Primitive nonprincipal finite-order character with conductor norm Q ; $-1/10 \leq \operatorname{Re} s \leq 11/10$. For principal characters use the regularized zeta function.	The small-row numerator is on $1/2$ with conductor $\ll_{\mathcal{A}} U$. Plain reflection uses its primitive functional equation and restores all redundant radicals.
Lemma 4.9	$a \in [1/2, 1]$, $0 < e < 10^{-3}$; zero-free disk of radius $2 - a - 2e$ about $2 + it$, or global $b \geq \beta^*$ and $\operatorname{Re} s \geq b + v$, $v > 0$.	Buffered disks are supplied by Lemma 8.1. Global reciprocals use $\beta^* + e$ or $\beta^* + 20e$. The extended prime moment explicitly uses $b = (1 + \kappa)/2 \geq \beta^*$.
Lemma 4.10	Squarefree deletion radical, local coefficients of modulus at most one; fixed positive real lower bound for inverse deletions; bounded strips for upper deletions.	Conductors and deletion supports are tracked separately and are polynomial. The primitive conjugate numerator follows from its original presentation and inverse deletions.
Lemma 5.2	Nonzero element rows, fixed finite S -supported ray family, fixed unit and S -valuation sectors; good moving primes remain moving.	The completed norm keeps rows meeting S , fixes sectors, and uses $f = 1$, puncture one.
Proposition 5.1; Lemmas 5.3–5.4; (5.27); (9.15)	Original completed cubic coefficients, fixed reflection modulus, genuine common kernel and original dual support; squarefree sieve indices and row-independent coefficients; original dual indices retain permitted S -primes.	Appendix A uses them in local reflection and the hybrid norm. The hybrid cubic column may contain permitted S -primes, as [1, (14.6), p.96] explicitly allows.
Lemma 7.1; (7.6), (7.11)–(7.13), (7.17)	Nonzero sixth-power-free u , $(u, S) = 1$, $\mathfrak{p} \notin S$, $j = 0, \dots, 5$, original unit phases and zero extensions.	All six cases are displayed in Section 4. New bounds are derived from these exact formulas.
Lemma 7.1, (7.14)	$s_r \geq 51/100$, $z_r \geq 17/50$, $w_r \geq -1/100$, $s_r + w_r \geq 1 + \varepsilon_0$, fixed $\varepsilon_0 > 0$.	Moderate paths use $10e$, small paths use $1/3$, and large paths lie farther right. Lemma 4.1 replaces the second-domain bound.
(7.2)	Nonnegative nonzero annular W_0 ; $M(z) = \int_0^\infty \widetilde{W}_0(r) r^{z-1} dr$ is holomorphic for $\operatorname{Re} z > 0$, with $M(\sigma) > 0$ for $0 < \sigma < 1$.	Section 6 uses $M_0(1/6) > 0$ and the nonnegative nonzero W_1 , giving $c_S > 0$ in (6.10).
§10.1	Fixed S , the original physical mask $(u, S) = 1$ and sixth-power-free rows.	Section 6 uses that $u = 1$ is the only principal numerator row. The other unit rows remain in the small-row range.

Input in [1]	Hypotheses retained	Application and verification
Lemmas 8.1–8.2; Proposition 8.3	The family $\mathcal{X}_u$, nonprincipal inducing rows, bounded d, r, m , $T_1 \leq U^{1/100}$, smooth annuli, common pure twists, one frequency allowance, and the stated height condition.	Section 2 records the conclusions. Theorem 5.1 checks the coefficient spans. Section 6 chooses τ after internal orders, including the pre-saturation range $[0, 22]$.
Lemma 10.2	Fixed compact real boxes; joint density with integrated and pointwise trace decay; arithmetic factor $Z^B(1 + \mathfrak{t})^J$ with B, J fixed before the tail order.	Equation (6.7) uses the original external smooth kernels. Buffered axes stay restricted. Internal measures with only weighted L^1 control are not used for traces.
Lemma 11.1	$1/2 < \sigma < 1$; common positive pre-height saving; finite $A_\eta, B_\eta, \tau_{0,\eta}$ before N ; J, f independent of T_1 .	Equation (6.21) supplies this order. Choose τ , then N , then the threshold.
Proposition 11.3	$1/2 \leq \sigma < 1$ and the whole primitive Hecke family zero-free in $\text{Re } s > \sigma$.	Every boundary here is in this range. Finite deletions are nonzero for $\text{Re } s > 0$; Section 2 handles pole cancellation at one.

Part II local input	Hypotheses retained	Why they hold here
Lemma 13.1	Fixed ray quotient, annular test, positive scale tending to infinity, fixed finite deletions.	Each $P_i = Z^{\ell_i}$ has $\ell_i > 0$ and its nonnegative bump is fixed before the target.
Lemmas 13.2–13.4	Complete finite moduli, original principal zero convention, all common factors extracted; CRT on genuinely coprime residual products.	The transforms keep full products, shared-prime multiplicities, roots, masks, and kernels before separation.
Corollary 14.1; Lemmas 14.2–14.3	Fixed finite-ray coefficients and row-independent puncture; disjoint product-form lists; squarefree hybrid indices; bounded lengths; row restrictions independent of active slots and duals apart from the exact zero mask.	Theorem 3.2 displays the actual coefficient and joint profile. The frozen rescaled tuple determines the row ball. The canonical terminal retains its allowed puncture and label.
Proposition 15.2	Original additive ray coefficients, fixed ball, polynomial $Q, Y' \geq 1$, and $P_a = Y'^2/Q \geq 1$.	The low proof gives the actual scales. Its inequality $l_y - \ell - 11b/6 > 0$ bounds the last Gram term by $P_a^{1/6}$.
Proposition 16.1	$51/100 \leq a \leq 1$, central contour, nonprincipal numerator, primitive conjugate bound (6.12), nonzero central selected factors.	The conjugation-closed family and common buffer give the correct primitive bound. Lemma 4.3 supplies the cutoff. Apply the strict-prime conductor saving once.
Lemma 17.1	Fixed finite-ray presentation, either common orientation, bounded row-independent prime coefficients, disjoint supports; $r + 2z \leq m - c_1$, $2r + 8z \leq 3m - c_2$, fixed $c_1, c_2 > 0$.	At width one, Theorem 5.1 gives $c_1 \geq 2\nu_0$ and $c_2 \geq 8\nu_0 + 9/37$. Whole conjugations preserve all zeros.
Lemma 17.2	Fixed character and puncture independent of row and label; squarefree divisor-bounded label weight depending only on that label; product-form mark; two strict puncture and width conditions.	Appendix A states these conditions and the initialization margins. No arbitrary extra column coefficient is introduced.

Part II local input	Hypotheses retained	Why they hold here
Lemma 17.3; Corollary 17.4	Finite acyclic graph, common coefficient measure, fixed exponents and height maps, finite weighted moments.	Both inductions strictly decrease width and retain common full-product profiles. Internal orders are chosen backwards.
Lemma 17.5	Primitive finite character with zero extension, complete mask, radial Schwartz profile, positive row scale; principal restoration on nonempty scale at least one.	The canonical transforms use genuine masks and untruncated divisor sums. Restored principal frequencies keep the outer mask.
Lemma 17.6	Sixth-power-free rows, original inverse coefficients and zeros; $U, D \geq 1$, $H = \max(2U, D^{1+c})$, $c > 0$; bounded real profile powers and rowwise twists.	These are the physical rows. The long and zero-capacity branches use small fixed c , including through $r = 1$.
Lemma 18.1, printed range	Same row and mask in both plains and slots; Θ coefficient span; inducing character outside Θ for positive slots; fixed-count-independent mesh; $n_1 + n_2 + 6\kappa z \leq M$ and the stated zero-free premise.	The original range is used at $\kappa = 3/4$, or at 1 for zero slots. Theorem 9.1 proves the extension. The cubic stage invokes it with $\kappa_a = 2\beta^* - 1$.
Lemma 18.2	Full common coefficient, complete Möbius allocations, natural zeros and whole-product phases, equal-product formal rectangles.	Every extraction in Section 9 retains that class and does not repuncture divisor quotients.
Lemma 18.3	Fixed character, common polynomial-norm squarefree mask, same norm power within a side, annular profiles, equal-product formal scales; all-scale fallback when lower scales fail.	Exceptional children stay formal rather than clipped and have the same coefficients and masks in both rectangles.
Lemma 19.1	Lemma 8.2's buffer and heights; prime-annulus frequency in the same allowance; original identity-ray slots.	Section 5 uses that class. Amplitude subdivision follows the original-bin contour move.
(12.5), (16.2)–(16.4), (16.7)	Distinct finite slot tuples, original positive windows and masks, arbitrary positive scales; marked coefficient $\eta(\mathfrak{p})$.	Equations (3.1) and (4.7) identify the same finite probe. Full quotient-free continuation is proved here.
Lemma 20.2, algebra only	Original numerical function, $0 \leq \delta \leq 5/6$, $0 \leq x \leq 1/2$, positive J .	Equations (7.1)–(7.2) reproduce its identity and signs. The changed geometry has an explicit increment.

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Use of AI

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